Previous Page

133. Y. Doucet and M. Bizouard, *Compt. Rend.*, 1960, **250**, 73.
134. L. F. Grantham, *J. Chem. Phys.*, 1965, **43**, 1415.
135. G. J. Janz and M. R. Lorenz, *J. Electrochem. Soc.*, 1961, **108**, 1052.
136. H. Bloom and E. Heymann, *Proc. Roy. Soc. (London)*, 1947, *A***188**, 392.
137. M. F. Lantratov and O. F. Moiseeva, *Zh. Fiz. Khim.*, 1960, **34**, 367.
138. L. A. King and F. R. Duke, *J. Electrochem. Soc.*, 1964, **111**, 712.

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS

The electrical conductivity in Ω^{-1} cm^{-1} at temperature t(°C) and composition p (wt.%) of the first-named constituent is given as κ_t, or empirical equations for κ as a function of temperature within the stated range are given at each composition. Principal references are in **bold type**.

AgBr–AgCl	p	0	24.7	46.7	66.9	84.0	100	
Ref. **59, 3**	κ_{450}	—	3.52	3.33	3.16	3.00	2.89	
	κ_{500}	3.90	3.70	3.50	3.31	3.16	3.02	
	κ_{550}	4.07	3.86	3.64	3.44	3.28	3.13	
	κ_{600}	4.21	3.99	3.77	3.55	3.38	3.21	
AgBr–KBr	p	52.2	61.9	78.0	90.1	100		
Ref. **59**	κ_{400}	—	—	1.42	1.98	—		
	κ_{500}	—	1.29	1.70	2.23	3.03		
	κ_{600}	1.43	1.55	1.91	2.43	3.21		
AgBr–KCl	p	38.6	62.6	79.0	91.0	100.0		
Ref. **77**, 78	κ_{500}	—	—	1.678	2.210	3.020		
	κ_{600}	—	1.680	1.942	2.432	3.215		
	κ_{700}	1.872	1.933	2.172	2.602	3.374		
	κ_{800}	2.127	2.150	2.340	2.742	3.505		
AgCl–$AgNO_3$	p	0	9.00	12.5	15.5	20.0	30.0	40.0
Ref. **36**	κ_{220}	0.706	0.705	0.710	0.712	—	—	—
	κ_{270}	0.926	0.926	0.931	0.932	0.950	1.005	1.092
	κ_{320}	1.132	1.132	1.138	1.138	1.161	1.223	1.312
AgCl–$AlBr_3$	p				13.29			
Ref. **13**	t	100	110	125	130	140	155	170
	κ	1.07	1.21	1.61	1.68	1.90	2.67	3.22
AgCl–KBr	p	23.2	44.6	64.4	83.0	100.0		
Ref. **77**, 78	κ_{500}	—	—	1.742	2.600	3.965		
	κ_{600}	1.400	1.570	2.008	2.871	4.277		
	κ_{700}	1.637	1.805	2.215	3.065	4.526		
	κ_{800}	1.815	1.980	2.390	3.220	4.716		
AgCl–KCl	p	50.2	63.0	72.5	87.9	100		
Ref. **59**	κ_{400}	—	—	—	2.30	—		
	κ_{500}	—	—	1.93	2.70	3.90		
	κ_{600}	—	2.01	2.24	3.00	4.21		
	κ_{700}	2.16	2.31	2.51	3.22	4.46		
AgCl–$PbCl_2$	p	0	11.2	25.6	43.5	76.3	100	
Ref. **59**	κ_{400}	—	—	1.27	1.67	2.62	—	
	κ_{500}	1.43	1.59	1.82	2.20	3.11	3.90	
	κ_{600}	1.92	2.07	2.27	2.61	3.46	4.21	
	κ_{700}	2.33	2.51	2.68	2.96	3.73	4.46	
AgCl–TlCl	p	0	20.6	37.0	58.0	77.2	100	
Ref. **3**	κ_{500}	1.215	1.470	1.711	2.260	2.925	3.653	
AgI–$AgNO_3$	p	0	10	20	30	40	50	60
Ref. **62, 33**	κ_{150}	—	—	—	0.38	0.44	0.54	0.65
	κ_{200}	—	—	0.58	0.60	0.66	0.75	0.86
	κ_{250}	0.80	0.78	0.78	0.80	0.85	0.93	1.03
	κ_{300}	1.00	0.98	0.98	1.00	1.05	1.12	1.21
$AgNO_3$–$Cd(NO_3)_2$	p	23.5	32.3	41.7	51.8	62.6	74.1	100
Ref. **46**	κ_{150}	—	—	0.07	0.15	—	—	—
	κ_{200}	—	0.11	0.19	0.28	0.38	0.54	—
	κ_{250}	0.15	0.23	0.34	0.44	0.55	0.65	0.83
	κ_{300}	0.27	0.38	0.49	0.61	0.73	0.86	—

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$AgNO_3$–HgI_2	p	4.0	13.8	27.2	35.9	46.6	59.9	77.0
Ref. **53, 19**	κ_{100}	—	—	—	0.08	0.07	0.08	0.08
	κ_{150}	0.02	0.11	0.12	0.20	0.15	0.22	0.31
	κ_{200}	0.09	0.24	0.22	0.30	0.27	0.36	0.51
$AgNO_3$–KNO_3	p	0	20	40	60	80	90	100
Ref. **71, 61, 51,** 79	κ_{250}	—	—	0.47	0.54	0.66	0.74	0.84
	κ_{300}	—	0.56	0.64	0.73	0.85	0.93	1.05
	κ_{350}	0.66	0.73	0.81	0.91	1.04	1.13	1.24
$AgNO_3$–$LiNO_3$	p	0	38.0	62.0	78.6	90.7	100	
Ref. **61**	κ_{250}	0.78	0.79	0.80	0.81	0.82	0.83	
	κ_{300}	1.05	1.04	1.04	1.04	1.04	1.04	
$AgNO_3$–$NaNO_3$	p	0	10	20	30	40	60	100
Ref. **38**	κ_{290}	—	0.900	0.907	0.910	0.920	0.941	1.010
	κ_{330}	1.076	1.088	1.094	1.104	1.103	1.118	1.173
	κ_{370}	1.251	1.267	1.264	1.277	1.272	1.289	—
$AgNO_3$–$RbNO_3$	p	22.3	33.0	43.4	63.3	72.9	82.2	91.2
Ref. **52**	κ_{200}	—	0.24	0.28	0.37	0.43	0.49	0.57
	κ_{250}	0.33	0.38	0.42	0.54	0.61	0.69	0.77
	κ_{300}	0.47	0.52	0.58	0.71	0.78	0.87	0.96
$AgNO_3$–$TlNO_3$	p	10	35	45	50	60	65	90
Ref. **29,** 3	κ_{100}	—	—	0.137	0.123	0.145	—	—
	κ_{200}	0.336	0.405	0.424	0.427	0.468	0.488	0.560
	κ_{300}	0.607	0.712	0.735	0.753	0.808	0.824	0.943
$AlBr_3$–$HgBr_2$	p	59.7	60.5	63.9	68.0	79.0	89.6	92.2
Ref. **25,** 3	$10^2 \cdot \kappa_{110}$	1.38	1.34	1.22	1.00	0.419	0.064	0.013
	$10^2 \cdot \kappa_{1\,140}$	2.45	2.34	2.00	1.61	0.573	0.064	0.013
$AlBr_3$–KBr	p	81.8	83.2	84.7	86.2	87.5	88.0	
Ref. **30,** 23, 2	κ_{110}	0.036 0	0.033 1	0.030 1	0.026 9	(0.024 2)	0.023 3	
	κ_{140}	0.060 1	0.054 6	0.049 3	0.043 9	(0.039 6)	0.037 9	
$2AlBr_3 \cdot KCl$	t	90	100	140	160	170		
Ref. **28,** 13	κ	0.023 2	0.029 5	0.058 7	0.074 2	0.082 9		
$4AlBr_3 \cdot LiCl$	t	80	100	120	140	170	180	
Ref. **13**	κ	0.011 0	0.013 9	0.017 0	0.020 4	0.024 2	0.025 1	
$AlBr_3$–$NaBr$	p	83.8	84.7	85.6	86.8	87.9	88.9	
Ref. **30**	κ_{120}	0.057 8	0.053 3	0.049 2	0.044 1	0.038 2	0.034 6	
	κ_{130}	0.067 4	0.061 3	0.056 5	0.050 6	0.044 4	0.040 2	
	κ_{140}	0.077 8	0.068 7	0.064 1	—	0.050 0	0.045 7	
$2AlBr_3 \cdot NaBr$	t	110	130	150	170			
Ref. **28**	κ	0.051 5	0.069 2	0.088 7	0.108 7			
$AlBr_3$–$NaCl$	p	90.73	93.69	94.83		at $p = 80.1$, $\kappa_{170} = 0.036\,5$		
Ref. **13**	κ_{110}	0.025 2	0.026 2	0.022 0				
	κ_{130}	0.041 5	0.031 0	0.025 2				
	κ_{150}	—	0.042 8	0.031 3				
$AlBr_3$–NH_4Br	p	84.50	86.90	88.98	89.42	90.81		
Ref. **25**	κ_{110}	0.022	0.019	0.017	0.016	0.014		
	κ_{130}	0.033	0.027	—	0.022	0.020		
	κ_{150}	0.043	0.036	0.031	0.029	0.025		
$2AlBr_3 \cdot NH_4Br$	t	110	160					
Ref. **28**	κ	0.022 0	0.048 4					
$AlBr_3$–$SbBr_3$	p	0	7.58	19.74	26.65	42.45	74.69	86.91
Ref. **23,** 2	$10^3 \cdot \kappa_{100}$	0	12	18	16	11	4	1
	$10^3 \cdot \kappa_{140}$	0	16	25	26	20	7	1
$AlBr_3 \cdot SbBr_3$	t	100	150	170				
Ref. **28**	$10^3 \cdot \kappa$	11.6	26.3	32.2				
$AlBr_3 \cdot SbBr_3$–$AsBr_3$	p	16.3	32.9	41.4	67.9	77.3	86.8	100
Ref. **35**	$10^2 \cdot \kappa_{85}$	0.089	0.121	0.252	0.766	0.855	0.892	0.803
	$10^2 \cdot \kappa_{100}$	0.082	0.116	0.264	0.938	1.125	1.228	1.155

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$AlBr_3$–$ZnBr_2$	p	73.43	78.03	82.57	87.03	91.42		
Ref. **23**	κ_{100}	5.5	5	3	1	0		
	κ_{130}	12.5	10.5	8	—	0		
	κ_{150}	20	17	11	2	0		
$2AlBr_3 \cdot ZnBr_2$	t	100	140	180				
Ref. **28**	$10^3 \cdot \kappa$	5.15	15.12	30.62				
$AlCl_3$–KCl	p	64.73	73.30	80.11	81.55	89.14		
Ref. **21**	κ_{200}	—	0.200	0.165	0.157	0.105		
	κ_{250}	0.346	0.280	0.228	0.214	0.143		
	κ_{300}	0.458	0.357	0.290	(0.270)	(0.180)		
$AlCl_3$–KCl**	p	50.4	53.7	62.5				
Ref. **31**	κ_{600}	0.790	0.811	0.858				
	κ_{700}	0.900	0.935	1.005				
	κ_{800}	1.010	1.060	1.155				
$AlCl_3 \cdot KCl$	t	250	300	500	600	700	800	
Ref. **31, 21**	κ	0.353	0.479	0.706	0.862	1.018	1.174	
$AlCl_3 \cdot LiCl$**	t	174	184	217	259	303.5	327	351
Ref. **21**	κ	0.354	0.380	0.468	0.553	0.647	0.687	0.731
$AlCl_3$–NaCl**	p	69.78	71.03	72.01	76.06	76.95	81.25	82.51
Ref. **15,** 14, 5	κ_{200}	0.436	0.262	0.280	0.160	0.170	0.090	0.070
	κ_{250}	0.532	0.378	0.380	—	0.300	0.178	—
$AlCl_3$–NH_4Cl	p	71.37	78.89	85.46	90.89			
Ref. **21**	κ_{200}	—	—	0.177	0.114			
	κ_{300}	0.479	0.387	0.302	0.194			
AlF_3–Na_3AlF_6*	p	0	2	4	6	8	10	20
Ref. **72, 65, 39,**	$\kappa_{1\,000}$	2.8	2.7	2.7	2.6	2.6	2.5	(2.3)
34, 12	$\kappa_{1\,050}$	2.9	2.9	2.8	2.8	2.7	2.6	(2.4)
Al_2O_3–Na_3AlF_6	p	0	3	6	9	12	15	20
Ref. **72, 65, 39, 34,**	$\kappa_{1\,000}$	2.8	2.6	2.5	2.3	2.2	2.1	(1.8)
80, 12, 4	$\kappa_{1\,100}$	3.0	2.9	2.7	2.6	2.4	2.3	(2.0)
Al_2O_3–Li_3AlF_6—	$p_{Al_2O_3}$	6.4	3.2	8.8	3.6	9.8	4.2	
Na_3AlF_6	$p_{Na_3AlF_6}$	46.8	58.2	64.0	67.6	72.2	76.6	
Ref. **80**	$\kappa_{1\,000}$	3.44	—	3.10	3.24	2.43	2.67	
	$10^3\alpha$	5.2	—	4.7	4.8	5.5	5.4	
	$\kappa_t = \kappa_{1\,000} + \alpha(t - 1\,000)$							
Al_2O_3–SiO_2	p	3	5.5	8	10	12		
Ref. **81**	a	−0.30	0.08	−0.40	−0.30	−1.00		
	$10^{-3}b$	5.9	5.7	4.1	4.8	3.0		
	$\log \kappa_t = a - b(t + 273)$ at 1 600–1 800							
B_2O_3–BaO	p	30.0	35.0	44.5	49.2	59.7	67.9	
Ref. **50**	κ_{900}	—	—	0.007	0.005	0.002	0.002	
	$\kappa_{1\,000}$	0.10	0.07	0.036	0.024	0.013	0.008	
	$\kappa_{1\,100}$	0.19	0.15	0.089	0.062	0.037	0.026	
	$\kappa_{1\,200}$	0.28	0.23	—	—	—	0.056	
B_2O_3–CaO	p	55.3	59.8	62.8	67.9	73.3		
Ref. **50**	$\kappa_{1\,000}$	—	—	—	0.025	0.013		
	$\kappa_{1\,100}$	—	0.12	0.10	0.069	0.047		
	$\kappa_{1\,200}$	0.23	0.22	0.20	0.151	0.100		
B_2O_3–K_2O	p	61.6	70.6	79.6	89.0	94.8	98.5	
Ref. **75, 42**	κ_{600}	0.016	0.003	—	1.8×10^{-4}	8.3×10^{-5}	1.1×10^{-5}	
	κ_{800}	0.186	0.056	0.022	6.0×10^{-3}	2.2×10^{-3}	2.2×10^{-4}	
	$\kappa_{1\,000}$	0.513	0.309	0.112	3.0×10^{-3}	9.1×10^{-3}	1.1×10^{-3}	
B_2O_3–Li_2O	p	79.9	87.1	92.0	95.5	97.2	98.5	
Ref. **75, 42**	κ_{700}	0.08	—	—	0.003	0.001	—	
	κ_{800}	0.87	0.10	0.03	0.011	0.004	0.001 2	
	κ_{900}	3.02	0.26	0.10	0.026	0.011	0.002 8	
	$\kappa_{1\,000}$	5.25	0.40	0.17	0.051	0.019	0.005 2	

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

B_2O_3–Na_2O	p	64.0	68.2	77.6	85.8	94.4	99.1	
Ref. **75, 42**	κ_{600}	—	—	—	6.5×10^{-4}	1.5×10^{-4}	1.6×10^{-5}	
	κ_{800}	—	0.33	0.08	0.022	4.8×10^{-3}	3.5×10^{-4}	
	κ_{1000}	1.74	1.32	0.40	0.107	2.1×10^{-2}	1.7×10^{-3}	
B_2O_3–PbO	p	88.2	92.7					
Ref. **69**	κ_{650}	5×10^{-7}	4×10^{-7}					
	κ_{750}	2×10^{-5}	0.6×10^{-5}					
	κ_{850}	8.5×10^{-5}	2.0×10^{-5}					
B_2O_3–SrO	p	44.9	49.6	56.8	61.6	65.2	70.0	
Ref. **50**	κ_{1000}	—	—	0.019	0.012	0.010	0.005	
	κ_{1100}	0.10	0.08	0.062	0.047	0.036	0.021	
	κ_{1200}	0.21	0.17	—	—	—	—	
$BaBr_2$–KBr	p	0	39.2	55.5	62.5	73.0	88.2	93.4
Ref. **43**	κ_{700}	—	1.17	1.03	0.85	0.89	—	—
	κ_{750}	1.63	1.29	1.16	0.96	1.00	0.93	—
	κ_{800}	1.76	1.41	1.27	1.06	1.13	1.06	0.98
	κ_{850}	1.87	1.52	1.39	1.17	1.25	1.19	1.12
$BaCl_2$–$CaCl_2$	p	17.3	32.0	44.6	55.6	65.3	73.8	88.3
Ref. **68**	κ_{800}	1.84	1.80	1.76	1.71	1.65	1.62	—
	κ_{900}	2.23	2.15	2.08	2.05	1.98	1.93	1.82
	κ_{1000}	2.57	2.51	2.44	2.38	2.32	2.27	2.14
	κ_{1100}	2.86	2.81	2.74	2.66	2.60	2.56	2.44
$BaCl_2$–$MgCl_2$†	p	13.4	31.0	49.7	73.2	75.7		
Ref. **37**	κ_{800}	1.405	1.620	1.70	1.61	1.57		
	κ_{900}	1.590	1.810	1.91	1.83	1.785		
	κ_{1000}	(1.775)	2.000	2.12	2.05	2.000		
$BaCl_2$–NaCl	p	28.4	47.1	59.8	70.4	78.1	84.2	93.4
Ref. **49, 17**	κ_{800}	3.00	2.55	2.26	2.00	1.54	1.20	—
	κ_{900}	3.26	2.90	2.68	2.50	2.32	2.25	2.00
	κ_{1000}	3.48	3.14	2.92	2.76	2.64	2.58	2.32
$BaCl_2$–$NaNO_3$	p	1.17	1.98	6.14	9.12			
Ref. **40**	κ_{350}	1.124	1.107	1.073	1.024			
	κ_{355}	1.143	1.128	1.093	1.052			
	κ_{360}	1.163	—	—	—			
BaF_2–NaF	p			66.6				
Ref. **11**	t	900	950	1 000	1 050	1 100		
	κ	4.027	4.335	4.602	5.051	5.319		
$Ba(NO_2)_2$–KNO_2	p	47.8	59.0	68.3	76.3	83.5	89.6	95.0
Ref. **82**	κ_{300}	—	0.494	0.438	0.393	—	—	—
	κ_{310}	—	0.528	0.472	0.428	0.381	—	0.311
	κ_{320}	—	0.560	0.505	0.464	0.415	0.386	0.345
	κ_{330}	—	0.594	0.542	0.498	0.450	0.418	0.377
	κ_{340}	0.670	0.628	0.572	0.534	0.486	0.452	0.411
	κ_{350}	0.702	0.660	0.610	0.570	0.520	0.484	0.445
$Ba(NO_2)_2$–KNO_3	p	26.8	31.3	43.8	54.8	64.5	73.3	81.0
Ref. **83**	κ_{320}	—	0.50	—	0.42	—	0.39	—
	κ_{340}	—	0.55	—	0.48	—	0.42	0.40
	κ_{360}	0.65	0.60	0.58	0.55	0.50	0.49	—
	κ_{380}	0.73	0.68	0.62	0.60	—	—	—
$Ba(NO_2)_2$–$NaNO_2$	p	19.4	39.2	53.9	64.0	73.9	84.5	94.5
Ref. **82**	κ_{240}	—	—	0.598	0.488	0.372	0.258	—
	κ_{260}	—	0.852	0.700	0.588	0.464	0.334	0.200
	κ_{280}	1.120	0.957	0.802	0.688	0.554	0.410	0.260
	κ_{300}	1.232	1.064	0.910	0.788	0.646	0.486	0.320
	κ_{320}	1.343	1.170	1.016	0.888	0.738	0.564	0.380
	κ_{340}	1.454	1.275	1.120	0.985	0.828	—	—
$Ba(NO_3)_2$–KNO_2	p	50.1	61.0	70.2				
Ref. **83**	κ_{260}	(0.38)	0.30	(0.28)				
	κ_{280}	0.41	0.36	(0.31)				
	κ_{300}	0.49	0.41	0.35				
	κ_{320}	0.55	0.46	0.40				

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$BeCl_2$–NaCl	p	39.1	45.2	49.2	58.3	63.6	70.8	83.2
Ref. **58**	κ_{300}	—	—	0.69	0.58	0.49	—	—
	κ_{400}	—	1.00	1.10	0.98	0.85	0.56	0.19
	κ_{500}	1.35	1.28	1.38	—	—	—	—
Bi–BiI_3	p	3.8	8.1	13.1	34.7	45.1	58.8	76.4
Ref. **84**	κ_{500}	0.5	0.8	2.1	200	600	1 400	3 100
Ca–$CaCl_2$	p	0.22	0.29	0.36	0.47	0.62	0.88	1.10
Ref. **85**	κ_{855}	2.50	2.52	2.55	2.58	2.61	2.68	2.73
$CaCl_2$–KCl	p	24.6	40.7	66.0	81.5			
Ref. **37,** 3	κ_{800}	1.770	1.500	1.415	1.560			
	κ_{900}	2.035	1.790	1.685	1.890			
$CaCl_2$–$MgCl_2$†	p	10.8	19.0	28.0	41.2	62.1	77.1	
Ref. **37**	κ_{700}	(1.18)	1.34	1.47	1.65	(1.79)	(1.87)	
	κ_{800}	1.37	1.53	1.68	1.90	2.07	2.20	
	κ_{900}	1.56	1.72	1.89	2.15	2.35	2.53	
$CaCl_2$–NaCl**	p	0	20	40	60	75	90	100
Ref. **67, 49, 17, 18,**	κ_{100}	—	2.78	2.31	1.92	1.71	—	—
8, 6, 3	κ_{800}	3.58	3.10	2.64	2.27	2.09	2.01	2.02
	κ_{900}	3.79	3.37	2.96	2.60	2.40	2.32	2.32
	$\kappa_{1\,000}$	4.00	3.58	3.22	2.92	2.71	2.67	2.68
CaF_2–CaO	p	90.7	92.6	94.5	96.4	98.2	100	
Ref. **55**	$\kappa_{1\,500}$	4.3	4.6	5.3	4.2	4.1	3.9	
	$\kappa_{1\,545}$	4.6	4.7	7.2	5.2	4.4	4.1	
CaF_2–Na_3AlF_6	p	0	5	10	20	30	40	
Ref. **65, 39, 26,**	$\kappa_{1\,000}$	2.8	2.8	2.7	2.6	2.5	2.4	
34, 12	$\kappa_{1\,100}$	3.0	3.0	3.0	2.9	2.8	2.7	
CaF_2–NaF	p	0	10					
Ref. **4,** 11	$\kappa_{1\,000}$	3.15	3.0					
	$\kappa_{1\,020}$	3.3	3.1					
$Ca(NO_3)_2$–KNO_3	p	0	16.9	24.0	34.2	44.8	54.9	65.5
Ref. **22**	κ_{300}	—	—	0.375	0.330	0.270	0.195	—
	κ_{350}	0.666	0.570	0.525	0.460	0.390	0.315	0.255
	κ_{400}	0.818	0.705	0.660	0.605	0.522	0.430	0.355
CaO–SiO_2	p	20	30	35	40	45	50	55
Ref. **81**	a	2.17	2.19	2.22	2.51	3.19	2.39	2.06
	$10^{-3}b$	5.7	5.7	5.2	5.7	6.7	5.0	4.1
				$\log \kappa_t = a - b/(t+273)$ at 1600–1800				
$CdBr_2$–$CdCl_2$	p	0	27.1	49.7	69.0	85.6	100	
Ref. **24**	κ_{600}	1.95	1.7	1.5	1.35	1.2	1.05	
	κ_{640}	2.1	1.85	1.65	1.5	1.35	1.25	
$CdCl_2$–CdI_2	p	0	14.2	33.2	59.9	100		
Ref. **43**	κ_{400}	0.22	0.35	—	—	—		
	κ_{500}	0.41	0.56	0.78	—	—		
	κ_{600}	0.66	0.79	1.03	1.32	1.96		
	κ_{700}	0.95	1.05	1.30	1.57	2.09		
$CdCl_2$–KCl	p	40	50	60	70	80	90	100
Ref. **47, 24, 3,** 27	κ_{500}	—	1.10	1.11	1.19	1.33	1.52	—
	κ_{600}	—	1.24	1.27	1.36	1.51	1.71	1.96
	κ_{700}	1.55	1.48	1.48	1.60	1.77	1.97	2.18
	κ_{800}	1.81	1.73	1.66	1.78	1.94	2.09	2.25
$CdCl_2$–LiCl	p	0	59.1	81.2	92.9	100		
Ref. **43**	κ_{550}	—	—	2.8	2.3	—		
	κ_{650}	6.0	4.1	3.2	2.6	2.1		
	κ_{750}	6.4	4.5	3.5	3.0	2.3		
$CdCl_2$–NaCl	p	43.9	67.6	82.5	92.6	100		
Ref. **24**	κ_{550}	(2.25)	1.95	1.9	1.85	1.8		
	κ_{600}	(2.4)	2.15	2.1	2.0	1.9		
	κ_{700}	2.7	2.45	2.4	2.3	2.2		

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$CdCl_2$–$PbCl_2$	p	0	20	40	60	80	100	
Ref. **56, 27, 24**	κ_{500}	1.44	1.68	—	—	—	—	
	κ_{600}	1.88	1.94	1.97	1.97	1.94	1.88	
	κ_{700}	2.28	2.35	2.36	2.31	2.26	2.20	
$CdCl_2$–TlCl	p	0	10	25	40	60	80	100
Ref.**44, 3**	κ_{500}	1.35	1.30	1.12	1.24	1.41	—	—
	κ_{600}	1.70	1.66	1.56	1.52	1.66	1.81	1.97
	κ_{700}	1.95	1.86	1.76	1.72	1.86	1.99	2.10
CdI_2–KI	p	0	19.7	35.5	64.3	75.3	89.8	100
Ref. **43**	κ_{300}	—	—	—	0.30	0.31	—	—
	κ_{500}	—	—	—	0.66	0.64	0.57	0.41
	κ_{700}	1.32	1.21	1.09	1.06	0.99	0.89	0.95
	κ_{800}	1.52	1.45	1.29	1.26	—	—	—
$Cd(NO_3)_2$–$CsNO_3$	p	39.4	44.6	54.8	64.5	73.8	82.9	
Ref. **45**	κ_{200}	—	—	0.09	0.07	—	—	
	κ_{250}	—	0.17	0.16	0.15	0.12	—	
	κ_{300}	0.26	0.25	0.24	0.23	0.20	0.15	
$Cd(NO_3)_2$–KNO_3	p	36.9	50.0	60.9	70.0	77.8	84.4	
Ref. **46**	κ_{200}	—	—	0.14	0.12	0.08	0.04	
	κ_{250}	—	—	0.26	0.23	0.18	0.12	
	κ_{300}	0.46	0.44	0.39	0.35	0.31	0.23	
$Cd(NO_3)_2$–$RbNO_3$	p	15.1	28.6	40.8	51.7	61.6	70.6	78.9
Ref. **46**	κ_{200}	—	—	0.12	0.10	0.10	0.08	—
	κ_{250}	—	0.22	0.21	0.20	0.19	0.16	—
	κ_{300}	0.31	0.34	0.32	0.31	0.29	0.26	0.20
$Cd(NO_3)_2$–$TlNO_3$	p	9.0	18.1	27.5	37.1	47.0	57.1	67.4
Ref. **45**	κ_{150}	—	—	0.14	0.11	0.08	—	—
	κ_{200}	—	0.29	0.25	0.22	0.17	—	—
	κ_{250}	0.47	0.31	0.38	0.34	0.28	0.23	—
	κ_{300}	0.60	0.54	0.50	0.46	0.40	0.34	0.26
Ce–$CeCl_3$	p	1.05	1.8	2.8	3.4	3.9	4.7	5.3
Ref. **95,** 96	κ_{855}	1.56	2.02	2.59	3.26	3.81	4.45	5.35
Ce–CeI_3	p	0.55	1.03	1.43	2.07	2.87	3.7	6.1
Ref. **86**	κ_{820}	1.00	1.92	2.74	4.75	8.88	15.5	*c.* 60
CsBr–NaCl	p	8.3	19.5	52.1	59.2			
Ref. **87**	κ_{800}	2.40	1.69	1.40	1.21			
	κ_{850}	2.55	1.79	1.50	1.30			
CsBr–$ZnSO_4$	p	39.8	63.8	79.9				
Ref. **62**	κ_{505}	0.03	0.09	0.20				
	κ_{550}	0.05	0.13	0.26				
CsCl–NaBr	p	29.4	52.2	71.0	87.0			
Ref. **87**	κ_{800}	2.05	1.61	1.50	1.45			
	κ_{850}	2.15	1.70	1.58	1.55			
CsCl–$TiCl_3$	p	52.1	62.3	72.0	81.5	90.8		
Ref. **88**	κ_{800}	0.7	0.7	0.8	0.9	1.2		
CuCl–KCl	p	25.8	42.0	47.6	57.5	76.0	85.9	100
Ref. **47**	κ_{450}	—	2.00	2.08	2.23	2.71	2.93	3.32
	κ_{500}	—	2.07	2.15	2.32	2.82	3.06	3.43
	κ_{550}	—	2.18	2.26	2.44	2.94	3.20	3.54
	κ_{600}	2.22	2.32	2.40	2.59	3.08	3.34	3.66
CuO–V_2O_5	p	10	30	40	50			
Ref. **74**	κ_{800}	0.6	3.2	3.8	5.5			
	κ_{900}	1.4	5.0	6.0	10.0			
	$\kappa_{1\,000}$	2.2	—	—	—			
Cu_2S–FeS	p	0	25	35	50	65	75	100
Ref. **60**	$\kappa_{1\,100}$	—	9.30	8.20	4.60	3.30	2.20	0.50
	$\kappa_{1\,300}$	1.530	9.40	8.30	5.20	4.60	3.80	1.30
	$\kappa_{1\,500}$	1.490	9.50	8.30	—	—	—	1.70

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$Dy–DyCl_3$	p	0.0	0.9	2.7	4.9	7.8	12.0	16.7
Ref. **89**	κ_{700}	0.42	0.46	0.54	0.65	0.76	0.90	0.98
$Er–ErCl_3$	p	0.0	1.0	3.0	3.8			
Ref. **89**	κ_{820}	0.50	0.55	0.67	0.69			
$FeO–SiO_2$	p	55.9	63.3	68.6	72.4	75.4	78.7	84.9
Ref. **41**	κ_{1300}	0.5	0.9	1.4	2.1	3.2	4.2	9.0
	κ_{1400}	0.8	1.6	2.3	4.0	5.0	7.8	—
$FeO–TiO_2$	p	32.6	35.7	56.5	62.1	91.9	97.2	
Ref. **63**	κ_{1300}	48	50	40	38	145	—	
	κ_{1400}	48	47	34	33	155	225	
$Fe_2O_3–V_2O_5$	p	15.0	19.4	30.5				
Ref. **74**	κ_{900}	1.4	3.8	6.8				
	κ_{1000}	2.0	5.8	11.6				
	κ_{1100}	3.6	10.6	—				
$Gd–GdBr_3$	p	0.0	0.4	1.3	2.3	2.7		
Ref. **89**	κ_{820}	0.47	0.58	0.87	1.27	1.46		
$Gd–GdCl_3$	p	0.0	0.3	0.6	1.0			
Ref. **89**	κ_{650}	0.41	0.43	0.45	0.49			
$HgBr_2–NH_4Br$	p	61.19	78.63	84.66	89.57	93.64	100	
Ref. **32**	κ_{250}	0.4	0.5	0.6(2)	0.6	0.45	0	
	κ_{300}	0.55	0.7	0.8(2)	0.75	0.6	0	
$HgCl_2–HgI_2$	p	0	20	40	50	60	80	100
Ref. **19**	κ_{225}	—	0.029 5	0.016 0	0.009 4	0.005 3	—	—
	κ_{250}	—	0.028 1	0.016 0	0.009 7	0.005 6	0.001 1	—
	κ_{300}	0.028 0	0.025 2	0.016 0	0.010 4	0.006 3	0.002 0	0.000 8
$HgCl_2–NH_4Cl$	p	71.42	81.82	83.54	86.12	92.21	93.84	95.31
Ref. **32**	κ_{250}	0.7	0.9	0.7	0.8	0.55	0.78	0.4
	κ_{300}	0.95	1.15	1.1	1.15	0.75	1.0	0.6
$HgCl_2–TlNO_3$	p	0	10	20	40	50	60	75
Ref. **19**	κ_{225}	0.416	0.344	0.274	0.184	0.150	0.126	—
	κ_{250}	0.476	0.397	0.320	0.226	0.186	0.156	0.100
	κ_{275}	0.536	0.450	0.368	0.269	0.222	0.188	0.126
$HgI_2–KI$	p	73.3	80.4	86.5	91.6	96.1	100	
Ref. **44**	κ_{270}	—	0.48	0.64	0.64	0.43	0.03	
	κ_{290}	0.50	0.56	0.69	0.69	0.45	0.03	
	κ_{310}	0.56	0.64	0.75	0.75	—	—	
$HgI_2–NH_4I$	p	67.64	75.82	82.47	87.98	92.62	96.58	100
Ref. **32**	κ_{250}	0.3	0.4	0.5	0.6	0.55	0.37	0.03
	κ_{350}	0.6	0.5	0.73	0.77	0.7	0.4	0.03
$HgI_2–TlNO_3$	p	0	20	40	50	60	80	100
Ref. **19**	κ_{250}	0.476	0.302	0.174	0.154	0.136	0.142	—
	κ_{275}	0.536	0.348	0.210	0.182	0.161	0.156	0.030 5
	κ_{300}	0.596	0.395	0.248	0.214	0.187	0.171	0.028 0
$Ho–HoCl_3$	p	0.0	0.7	3.2	6.3	10.6		
Ref. **89**	κ_{800}	0.53	0.56	0.69	0.84	1.05		
K–KBr	p	1	2	3	4	6	8	
Ref. **73**	κ_{763}	5.6	11.7	22.4	40.8	104.7	229.2	
	κ_{870}	6.8	16.3	29.5	51.3	129.0	263.0	
K–KCl	p	1	2	4	6	8	10	
Ref. **73**	κ_{816}	3.8	6.2	14.1	31.6	64.6	117.5	
K–KF	p	1.35	2.72	4.10	4.80			
Ref. **90**	κ_{900}	4.5	6.7	12.0	18.0			
K–KI	p	0.5	1.0	1.5	2.0	—	—	—
Ref. **90, 91**	κ_{900}	4.5	8.5	13.5	19.0	—	—	—
	p	0.5	1.2	2.6	13.5	26.0	48.7	64.9
	κ_{700}	3.7	7.9	19.7	500	1 500	4 200	7 300
$KBF_4–KF$	p	30	40	50	60	70	80	90
Ref. **92**	κ_{600}	—	—	—	—	1.0	2.0	1.5
	κ_{700}	—	—	3.0	5.3	6.8	7.2	6.7
	κ_{800}	5.0	8.0	10.0	11.5	12.5	12.8	12.6

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

KBr–KNO_3	p	0	17.2	37.0	54.1	70.2	86.9	100
Ref. **43**	κ_{400}	0.82	0.79	0.78	—	—	—	—
	κ_{600}	1.38	1.34	1.29	1.30	1.31	—	—
	κ_{700}	—	—	—	—	1.53	1.53	—
	κ_{800}	—	—	—	—	1.76	1.76	1.76
KBr–NaCl	p	33.7	57.6	85.9	89.0			
Ref. **93**	κ_{800}	2.97	2.51	2.20	1.95			
	κ_{850}	3.05	2.60	2.29	2.05			
KBr–RbCl	p	19.7	39.5	59.5	81.3			
Ref. **94**	κ_{770}	1.64	1.64	1.64	1.65			
	κ_{800}	1.70	1.70	1.70	1.72			
	κ_{850}	1.80	1.80	1.80	1.82			
KBr–TlBr	p	2.2	5.3	9.7	18.4	30.3	53.9	72.0
Ref. **97**	a	0.489_1	0.441_9	2.008_7	2.390_4	0.475_4	0.808_8	0.426_5
	$10^3 b$	3.239_7	2.957_2	7.546_5	8.864_5	3.284_7	4.267_7	5.332_0
	$10^6 c$	0.835_7	0.499_1	3.845_7	4.904_9	0.845_0	1.478_0	0.865_2
	t range	537–701	473–640	650–762	619–717	626–791	687–796	718–783
				$\kappa_t = a + bt - ct^2$				
KBr–$ZnSO_4$	P	35.8	40.0	44.4	50.2	60.0	64.7	73.0
Ref. **48**	κ_{475}	0.10	—	—	0.13	0.21	0.24	0.34
	κ_{500}	0.12	0.15	0.14	0.16	0.25	0.29	0.41
	κ_{550}	0.16	0.21	0.19	0.23	0.33	0.39	0.55
KCl–KI	p	2.8	7.8	13.3	26.8	41.2	64.4	
Ref. **57**	κ_{700}	1.31	1.38	1.41	1.48	1.56	1.66	
	κ_{800}	1.52	1.56	1.61	1.70	1.80	1.96	
	κ_{900}	1.66	1.72	1.76	1.88	1.97	2.14	
KCl–LiCl**	p	0	28.2	42.6	55.8	72.2	87.6	100
Ref. **57, 10**	κ_{500}	—	—	2.31	1.87	—	—	—
	κ_{600}	—	3.51	2.87	2.40	1.90	—	—
	κ_{700}	6.17	4.10	3.42	2.88	2.35	2.04	—
	κ_{800}	6.62	4.56	3.82	3.24	2.72	2.36	2.24
KCl–$MgCl_2$	p	10	20	30	50	60	70	80
Ref. **56, 37, 16,**	κ_{600}	—	—	0.97	0.98	0.97	0.98	—
9, 7	κ_{700}	1.10	1.17	1.21	1.24	1.27	1.34	1.48
	κ_{800}	1.33	1.37	1.39	1.44	1.48	1.56	1.76
	κ_{900}	1.50	1.54	1.56	1.62	1.67	1.75	1.87
KCl–NaBr	p	22.0	32.0	51.4	74.0			
Ref. **93**	κ_{800}	2.64	2.40	2.29	2.21			
	κ_{850}	2.76	2.53	2.39	2.32			
KCl–NaCl**	p	18.7	32.1	40.6	54.8	64.8	82.9	
Ref. **57, 16,** 9, 6, 3	κ_{700}	—	—	2.52	2.35	2.21	—	
	κ_{800}	3.23	2.98	2.85	2.64	2.53	2.34	
	κ_{900}	3.50	3.24	3.11	2.88	2.76	2.56	
KCl–NaI	p	0	8.0	19.8	33.0	49.7	73.6	100
Ref. **43**	κ_{600}	—	1.88	1.68	1.54	1.51	—	—
	κ_{700}	2.41	2.20	2.04	1.86	1.84	1.87	—
	κ_{800}	2.68	2.51	2.37	2.16	2.44	2.17	2.22
KCl–$PbCl_2$	p	2.8	4.5	6.1	14.8	20.7	37.9	51.2
Ref. **98,** 47, 27, 24	κ_{450}	—	1.176	1.142	1.034	0.937	—	—
	κ_{500}	1.414	1.401	1.345	1.205	1.106	1.038	(1.110)
	κ_{550}	1.645	1.609	1.558	1.364	1.273	1.174	1.230
	κ_{600}	1.868	1.807	1.723	1.515	1.417	1.306	1.350
	κ_{650}	2.086	2.018	1.913	1.684	1.572	1.467	1.519
	κ_{700}	—	—	—	—	1.716	1.624	1.670
KCl–RbBr	p	9.9	22.7	39.8	63.9			
Ref. **94**	κ_{770}	1.42	1.58	1.70	1.88			
	κ_{800}	1.50	1.65	1.77	1.98			
	κ_{850}	1.59	1.73	1.89	2.09			

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$KCl–SnCl_2$	p	0.0	4.1	8.7	14.1	20.3	27.8	31.9
Ref. **99**	κ_{300}	1.115	1.079	0.995	0.865	0.688	0.472	—
	κ_{350}	1.388	1.331	1.215	1.060	0.871	0.645	0.702
$KCl–TiCl_3$	p	32.6	42.0	53.0	65.9	81.4		
Ref. **88**	κ_{800}	1.0	1.05	1.3	1.5	1.8		
$KCl–ZnCl_2$	p	0	10.5	17.8	26.3	35.9	52.6	58.7
Ref. **66**	κ_{500}	0.09	0.40	0.64	0.70	0.76	0.78	0.79
	κ_{550}	0.16	0.51	0.77	0.82	0.89	0.92	0.95
	κ_{600}	0.26	0.63	0.90	0.95	1.01	1.05	1.09
	κ_{650}	0.37	0.76	1.02	1.06	1.14	1.19	1.24
$KCl–ZnSO_4$	p	19.0	24.0	29.3	33.7	39.8	50.0	56.8
Ref. **48**	κ_{475}	—	0.09	0.13	0.16	0.21	0.30	0.36
	κ_{500}	0.09	0.12	0.17	0.21	0.25	0.35	0.41
	κ_{550}	0.14	0.16	0.24	0.27	0.32	0.44	0.51
$K_2Cr_2O_7–KNO_3$	p	0	24.4	42.1	55.5	74.4	87.1	100
Ref. **44**	κ_{350}	0.56	0.52	0.42	0.35	0.22	0.17	—
	κ_{400}	0.72	0.66	0.64	0.48	0.34	0.26	0.20
	κ_{475}	0.95	0.88	0.74	0.67	0.52	0.41	0.35
KF–NaCl	p	33.20	49.85	66.54	100			
Ref. **6**	κ_{750}	2.6	2.9	3.2	—			
	κ_{850}	3.4	3.6	3.7	4.1			
	κ_{950}	4.2	4.3	(4.4)	4.5			
KI–NaCl	p	33.6	55.0	74.1	86.9	94.2	100	
Ref. **43**	κ_{600}	—	—	1.56	1.35	—	—	
	κ_{700}	—	2.24	1.87	1.64	1.47	1.32	
	κ_{800}	3.06	2.58	2.20	1.91	1.73	1.52	
$KI–ZnSO_4$	p	47.4	50.9	54.8	60.4	62.9	67.3	
Ref. **48**	κ_{440}	—	—	0.11	0.13	0.15	0.21	
	κ_{460}	0.09	0.08	0.13	0.16	0.18	0.24	
	κ_{480}	0.12	0.10	0.15	0.18	0.20	0.28	
$K_2MoO_4–MoO_3$	p	17.5	27.5	43.0	52.2	60.0	79.4	91.0
Ref. **100**	a	−3.660 6	−1.312 0	−1.982 0	−1.208 3	−2.728 0	6.064 4	4.250 6
	10^2b	0.942 8	0.336 5	0.534 5	0.333 3	0.832 9	−1.299 9	−0.853 7
	10^6c	−5.010 1	−1.058 9	−2.442 6	−1.206 1	−5.307 3	8.173 1	5.507 5
	t range	760–869	614–804	654–783	642–794	583–705	882—934	877–995
				$\kappa_t = a + bt + ct^2$				
$KNO_2–NaNO_2$	p	12.0	23.6	32.7	45.1	55.3	74.2	83.3
Ref. **82**	κ_{300}	1.262	1.176	1.108	1.052	0.998	—	—
	κ_{340}	1.472	1.380	1.308	1.244	1.185	1.040	—
	κ_{380}	—	1.586	1.510	1.434	1.366	1.204	1.130
	κ_{420}	—	—	—	1.625	1.548	1.365	1.287
	κ_{460}	—	—	—	—	—	1.525	1.442
$KNO_3–LiNO_3$	p	66.0						
Ref. **101**	$\kappa_t = -0.378\,8 + 3.40 \times 10^{-3}t$ at 160–240							
	$\kappa_t = -0.403\,7 + 3.47 \times 10^{-3}t$ at 240–400							
$KNO_3–NaNO_3$**	p	0	20	40	50	60	80	100
Ref. **19**, 101, 79, 1	κ_{300}	—	0.820	0.706	0.652	0.625	0.572	—
	κ_{400}	1.364	1.160	1.024	0.966	0.931	0.874	0.818
	κ_{500}	1.720	1.496	1.351	1.284	1.234	1.178	1.107
$KNO_3–RbNO_3$	p	7.1	18.6	31.4	40.7	56.0	73.3	86.0
Ref. **45**	κ_{300}	—	0.40	0.42	—	—	—	—
	κ_{350}	0.51	0.52	0.55	0.57	0.59	0.63	0.64
	κ_{400}	0.63	0.65	0.68	0.70	0.73	0.77	0.79
$K_2SO_4–ZnSO_4$	p	31.6	41.8	52.0				
Ref. **62**	κ_{450}	0.04	0.06	0.05				
	κ_{500}	0.06	0.08	0.08				
	κ_{550}	0.07	0.12	0.12				
$K_2TaF_7–Ta_2O_5$	p	78.0	83.4	89.0	94.5	100.0		
Ref. **102**	κ_{800}	0.71	0.80	0.87	0.91	0.92		
	κ_{900}	0.85	0.94	1.00	1.04	1.05		

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

K_2TiF_6–TiO_2	p	81.9	87.5	92.3	94.5	96.6	98.3	100
Ref. **102**	κ_{870}	1.18	1.22	1.28	1.30	1.31	1.39	1.40
	κ_{920}	1.29	1.33	1.37	1.39	1.40	1.48	1.49
K_2WO_4–WO_3	p	37.4	50.5	58.3	67.4	76.6	85.0	92.2
Ref. **103**	a	2.815 6	2.880 9	2.311 1	2.356 2	5.493 6	7.543 6	38.80
	10^2b	0.560 1	0.615 0	0.549 2	0.565 0	1.250 1	1.640 2	7.964 6
	10^6c	2.087 5	2.501 1	2.432 4	2.392 3	5.882 2	7.524 8	39.636
	t range	891–1 010	759–912	754–770	674–819	773–955	842–1 016	885–1 029
				$\kappa_t = -a + bt - ct^2$				
La–$LaBr_3$	p	0.0	0.4	1.2	2.5	4.1	5.5	6.5
Ref. **89**	κ_{820}	0.70	0.94	1.55	3.15	7.35	15.8	25.9
La–$LaCl_3$	p	0.3	1.1	2.0	3.0	4.6	6.2	
Ref. **85**	κ_{910}	1.51	1.84	2.46	3.21	5.02	7.21	
La–LaI_3	p	0.5	1.4	2.4	3.5	4.4		
Ref. **86**	κ_{840}	0.87	2.39	5.54	12.2	20.9		
Li_3AlF_6–Na_3AlF_6	p	20	30	50	80			
Ref. **80**, 65	$\kappa_{1\,000}$	3.05	3.38	3.80	4.33			
	$10^3\alpha$	4.8	4.8	4.8	4.8			
		$\kappa_t = \kappa_{1\,000} + \alpha(t - 1\,000)$ at 762–936						
Li_2MoO_4–MoO_3	p	19.5	33.9	51.5	60.5	71.3	85.2	91.0
Ref. **100**	a	10.470 8	7.490 5	4.031 1	4.612 7	2.795 8	21.335 9	13.230 5
	10^2b	2.498 1	1.815 3	1.028 9	1.183 2	0.753 9	5.368 6	3.225 9
	10^6c	13.536	9.358 9	4.489 7	5.211 7	2.887	30.734	16.322
	t range	778–917	800–907	768–917	799–945	845–956	764–872	809–908
				$\kappa_t = -a + bt - ct^2$				
$LiNO_3$–$RbNO_3$	p	10.5	16.7	23.8	31.9	52.3	65.2	80.8
Ref. **52**	κ_{200}	—	0.20	0.24	—	—	—	—
	κ_{250}	—	0.33	0.36	0.41	0.53	0.61	—
	κ_{300}	0.43	0.47	0.51	0.56	0.72	0.82	0.96
Li_2WO_4–WO_3	p	54.6	59.7	69.0	77.5	85.2	92.3	
Ref. **103**	a	6.015 2	2.149 1	7.464 8	0.606 3	5.259 7	5.723 7	
	10^2b	1.333 6	0.455 8	1.764 2	0.108 6	1.277 3	1.404 8	
	10^6c	5.594 8	0.392 3	7.669 4	−1.989 8	4.823 7	5.797 8	
	t range	805–952	769–924	757–938	735–890	716–897	734–907	
				$\kappa_t = -a + bt - ct^2$				
$MgCl_2$–NaCl	p	23.2	39.1	57.2	69.0	81.5	90.3	
Ref. **37**, 56, 16, 9	κ_{700}	2.80	2.31	1.97	1.78	1.63	1.39	
	κ_{800}	2.96	2.49	2.16	1.98	1.82	1.57	
	κ_{900}	3.12	2.67	2.35	2.18	2.01	1.75	
MgF_2–Na_3AlF_6	p	5	10					
Ref. **72**	κ_{950}	—	2.2					
	κ_{975}	2.5	—					
MnO–SiO_2	p	40	50	55	60	65	70	80
Ref. **81**	a	2.32	2.26	1.85	2.25	2.94	2.21	2.00
	$10^{-3}b$	4.8	4.1	3.1	3.4	4.6	2.5	1.7
				$\log \kappa_t = a - b/(t + 273)$ at 1 600–1 800				
MoO_3–Na_2MoO_4	p	21.4	26.4	51.2	62.2	71.2	83.7	86.2
Ref. **54**	κ_{700}	1.06	1.07	0.86	0.80	0.75	0.63	—
	κ_{800}	1.45	1.41	1.17	1.11	1.08	0.75	0.91
	κ_{900}	—	—	—	—	—	—	1.16
Na–NaBr	p	0.25	0.5	0.75	1.0	1.5	2.0	
Ref. **73**	κ_{803}	4.0	6.3	7.4	8.2	—	—	
	κ_{894}	4.8	7.7	9.7	11.5	14.7	17.5	
Na–NaCl	p	0.25	0.5	0.75	1.0	1.5		
Ref. **73**	κ_{848}	4.3	4.9	5.3	5.7	6.4		
	κ_{893}	4.4	5.0	5.5	5.9	6.7		

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

Na–NaI	p	0.4	0.6	1.0	1.3	1.6	2.0	
Ref. **90**	κ_{700}	3.7	—	—	—	—	—	
	κ_{800}	4.5	3.5	6.5	—	—	—	
	κ_{900}	5.0	6.5	8.0	10.0	12.2	15.0	
Na_3AlF_6–NaF‡	p	0	20	40	60	80	90	100
Ref. **65, 39, 34,** 12	$\kappa_{1\,000}$	5.4	4.9	4.4	3.9	3.3	3.1	2.8
	$\kappa_{1\,050}$	5.7	5.1	4.6	4.0	3.5	3.2	2.9
	$\kappa_{1\,100}$	5.9	5.4	4.8	4.2	3.6	3.3	3.1
$NaBF_4$–NaF	p	40	50	60	70	80	90	
Ref. **92**	κ_{500}	—	—	1.5	3.0	3.5	4.0	
	κ_{600}	—	3.0	5.0	6.0	6.5	7.0	
	κ_{700}	—	5.0	8.0	10.0	10.5	10.5	
	κ_{800}	6.5	12.0	14.0	15.0	15.5	15.0	
NaBr–NaOH	p	17.7	25.0	30.4	39.8	48.4	52.3	
Ref. **70**	κ_{300}	1.18	1.14	1.03	0.95	0.83	0.79	
	κ_{400}	2.00	1.96	1.83	1.75	1.63	1.59	
NaCl–Na_2CO_3	p	0	52.45	68.81				
Ref. **6**	κ_{700}	—	2.2	2.3				
	κ_{850}	2.4	3.1	3.0				
	$\kappa_{1\,000}$	2.8	3.1	3.4				
NaCl–$NbCl_5$	p	43.8	60.4	60.7	71.8	73.6	100	
Ref. **20**	κ_{800}	1.32	2.20	2.19	—	—	3.34	
	κ_{850}	1.83	2.60	2.59	2.98	2.94	3.60	
NaCl–$PbCl_2$	p	0	2.55	9.01	20.05			
Ref. **17**	κ_{500}	1.445	1.51	1.58	1.645			
	κ_{600}	1.91	1.95	2.005	2.07			
NaCl–$TiCl_3$	p	27.5	36.2	47.0	60.3	77.4		
Ref. **88**	κ_{800}	1.75	1.75	1.9	2.3	2.7		
NaCl–$ZrCl_4$	p	11.5	34.9	59.7	75.1	88.1	88.7	
Ref. **20, 104**	κ_{350}	0.46	—	—	—	—	—	
	κ_{800}	—	1.45	1.74	2.41	2.88	2.68	
	κ_{825}	—	—	1.83	2.83	3.08	—	
	κ_{850}	—	—	1.97	3.01	3.34	3.36	
NaF–SrF_2	p	—	—	33.3	—	—		
Ref. **11**	t	900	950	1000	1050	1100		
	κ	4.441	4.798	4.961	5.407	5.642		
NaI–NaOH	p	21.5	35.0	44.8	52.8			
Ref. **70**	κ_{300}	1.14	0.87	0.72	0.63			
	κ_{400}	1.90	1.53	1.32	1.23			
$NaNO_2$–$NaNO_3$	p	0	21.3	44.8	65.4	82.1	100	
Ref. **43**	κ_{300}	—	1.02	1.11	1.19	1.25	1.34	
	κ_{350}	1.16	1.27	1.34	1.44	1.51	1.62	
	κ_{400}	1.36	1.51	1.58	1.69	1.78	1.89	
	κ_{450}	1.55	1.73	1.79	1.92	2.02	2.10	
Na_2O–TiO_2	p	28.3	41.7	44.6	51.7	60.0	65.5	71.6
Ref. **64**	$\kappa_{1\,000}$	—	0.32	0.38	0.47	0.56	0.70	0.86
	$\kappa_{1\,100}$	0.32	0.45	0.56	0.80	0.90	0.94	1.15
	$\kappa_{1\,200}$	0.74	1.03	1.16	1.55	1.76	2.10	2.54
Na_2WO_4–WO_3	p	45.9	50.9	55.7	65.1	74.2	83.0	91.6
Ref. **103**	a	4.450 9	7.001 3	4.563 4	7.275 4	3.136 8	6.074 3	2.748 1
	10^2b	0.962 5	1.547 6	1.032 6	1.720 1	0.739 9	1.550 6	0.671 8
	10^6c	4.036 4	7.032 0	4.455 5	8.560 0	2.422 0	7.194 3	2.495 8
	t range	780–923	750–888	767–905	741–881	689–850	653–803	701–869
				$\kappa_t = -a + bt - ct^2$				
Nd–$NdBr_3$	p	0.4	1.5	3.6	7.5	11.4	13.9	15.4
Ref. **89**	κ_{730}	0.53	0.61	0.79	0.94	1.01	0.96	0.92

(*continued*)

Table 9.5 ELECTRICAL CONDUCTIVITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$NdCl_3$–$NdCl_3$	p	1.7	2.4	3.8	7.3	11.0	14.6	18.6
Ref. **85**	κ_{855}	1.20	1.24	1.32	1.54	1.67	1.77	1.84
Nd–NdI_3	p	0.3	0.9	2.1	5.2	8.1	11.1	11.5
Ref. **86**	κ_{820}	0.476	0.585	0.759	1.033	1.102	0.980	0.930
$PbBr_2$–$PbCl_2$	p	0	23.7	45.3	66.6	83.8	100	
Ref. **59, 3**	κ_{450}	—	—	1.02	0.97	0.91	0.84	
	κ_{500}	1.43	1.34	1.25	1.17	1.10	1.03	
	κ_{550}	1.69	1.56	1.45	1.36	1.26	1.19	
PbO–SiO_2	p	61.4	71.2	78.8	84.8	89.7	93.7	97.1
Ref. **105**	$\log \kappa_{900}$	−2.9	−2.0	−1.2	−0.7	−0.3	0.0	0.2
	$\log \kappa_{1150}$	−1.9	−1.2	−0.6	−0.2	0.0	0.2	0.6
Pr–$PrBr_3$	p	0.5	1.1	2.5	4.1	5.8	8.6	8.7
Ref. **89**	κ_{740}	0.66	0.80	1.10	1.47	1.90	2.38	2.67
Pr–PrI_3	p	0.3	1.3	2.6	4.2	6.3	8.1	8.9
Ref. **86**	κ_{780}	0.61	1.28	2.48	4.24	8.25	14.8	21.0
RbCl–$TiCl_3$	p	43.9	54.0	64.6	75.8	87.6		
Ref. **88**	κ_{800}	0.95	0.95	1.0	1.2	1.4		
VCl_2 in KCl–NaCl	p	9.6	12.8	17.9	22.4	29.7	39.7	52.0
(equi-molar:	κ_{600}	0.22	0.48	2.34 (610)	1.30	1.18 (605)	1.46	0.050
56.1% KCl)	κ_{800}	3.68 (810)	3.77 (790)	3.94 (790)	3.94	3.85	3.68 (795)	2.93
Ref. **106**								

**See also* Na_3AlF_6–NaF.

***See also* 'Physical Properties Data Compilations Relevant to Energy Storage II Molten Salts: Data on single and Multi-Component Salt Systems', Janz *et al.*, NSRDS–NBS 61, Part II.

† Melts may contain up to 1% MgO.

‡ *See also* AlF_3–Na_3AlF_6.

REFERENCES TO TABLE 9.5

1. H. M. Goodwin and R. D. Mailey, *Phys. Rev.*, 1908, **26**, 28.
2. V. A. Izbekov and V. A. Plotnikov, J. *Russ. Phys. -Chem. Soc.*, 1911, **43**, 18.
3. C. Sandonnini, *Gazz. Chim. Ital.*, 1920, **50**, 289.
4. K. Arndt and W. Kalass, *Z. Elektrochem.*, 1924, **30**, 12.
5. V. A. Plotnikov and P. T. Kalita, *J. Russ. Phys. -Chem. Soc.*, 1930, **62**, 2195.
6. E. Ryschkewitsch, *Z. Elektrochem.*, 1933, **39**, 531.
7. S. V. Karpachev, A. G. Stromberg and O. Poltoratzkaya, *J. Phys. Chem. (USSR)*, 1934, **5**, 793.
8. V. P. Barzakovskii, Proc. 1st All-Union Conf. on Non-Aqueous Solutions, 1935, 153.
9. K. P. Batashev, *Metallurg. (USSR)*, 1935, **10**, 100.
10. S. V. Karpachev, A. G. Stromberg and V. N. Podchainova, *J. Gen. Chem. (USSR)*, 1935, **5**, 1517.
11. M. de Kay Thompson and A. L. Kaye, *Trans. Electrochem. Soc.*, 1935, **67**, 169.
12. K. P. Batashev, *Legkie Metal.*, 1936, **10**, 48, quoted by V. P. Mashovets, 'The Electrometallurgy of Aluminium', 1938.
13. Ya. P. Mezhennii, *Mém. Inst. Chem., Acad. Sci. Ukrain S.S.R.*, 1938, **4**, 413.
14. Y. Yamaguti and S. Sisido, *J. Chem. Soc. Japan*, 1938, **59**, 1311.
15. A. I. Kryagova, *J. Gen. Chem. (USSR)*, 1939, **9**, 2061.
16. A. A. Sherbakov and B. F. Markov, *J. Phys. Chem. (USSR)*, 1939, **13**, 621.
17. V. P. Barzakovskii. *Bull. Acad. Sci. U.R.S.S.*, Class Sci. Chim., 1940, 825.
18. V. P. Barzakovskii, *J. Appl. Chem. (USSR)*, 1940, **13**, 1117.
19. A. G. Bergman and I. M. Chagin, *Bull. Acad. Sci. U.R.S.S.*, Class Sci. Chim., 1940, 727.
20. N. A. Belozerskii and B. A. Freidlina, *J. Appl. Chem. (USSR)*, 1941, **14**, 466.
21. Y. Yamaguti and S. Sisido, *J. Chem. Soc. Japan*, 1941, **62**, 304.
22. E. R. Natsvilishvili and A. G. Bergman, *Bull Acad. Sci., U.R.S.S.*, Class Sci. Chim., 1943, 23.
23. E. Ya. Gorenbein, *J. Gen. Chem. (USSR)*, 1945, **15**, 729.
24. H. Bloom and E. Heymann, *Proc. R. Soc.*, 1947, **188***A*, 392.
25. E. Ya. Gorenbein, *J. Gen. Chem. (USSR)*, 1947, **17**, 873.
26. T. G. Pearson and J. Waddington, *Faraday Soc. Discussion*, 1947, No. 1, 307.
27. N. M. Tarasova, *J. Phys. Chem. (USSR)*, 1947, **21**, 825.
28. E. Ya. Gorenbein, *J. Gen. Chem. (USSR)*, 1948, **18**, 1427.
29. I. I. Bokhovkin, *ibid.*, 1949, **19**, 805.
30. E. Ya. Gorenbein and E. E. Kriss, *ibid.*, 1949, **19**, 1978.

31. H. Grothe, *Z. Elektrochem.*, 1949, **53**, 362.
32. I. N. Belyaev and K. E. Mironov, *Dokl. Akad. Nauk SSSR*, 1950, **73**, 1217.
33. I. I. Bokhovkin, *J. Gen. Chem. (USSR)*, 1950, **20**, 397.
34. A. Vayna, *Alluminio*, 1950, **19**, 215.
35. E. Ya. Gorenbein and E. E. Kriss, *J. Phys. Chem. (USSR)*, 1951, **25**, 791.
36. R. C. Spooner and F. E. W. Wetmore, *Canad. J. Chem.*, 1951, **29**, 777.
37. R. W. Huber, E. V. Potter and H. W. St. Clair, *Rep. Invest. US Bur. Mines, No. 4858*, 1952.
38. J. Byrne, H. Fleming and F. E. W. Wetmore, *Canad. J. Chem.*, 1952, **30**, 922.
39. J. D. Edwards, C. S. Taylor, L. A. Cosgrove and A. S. Russell, *Trans. Electrochem. Soc.*, 1953, **100**, 508.
40. A. Bogorodski, *J. Soc. Phys.-Chim. Russe*, 1905, **37**, 796.
41. K. Mori and Y. Matsushita, *Tetsu to Hagane*, 1952, **38**, 365.
42. L. Shartsis, W. Capps and S. Spinner, *J. Am. Ceram. Soc.*, 1953, **36**, 319.
43. H. Bloom, I. W. Knaggs, J. J. Molloy and D. Welch, *Trans. Faraday Soc.*, 1953, **49**, 1458.
44. I. N. Belyaev, *Invest. Sekt. Fiziko-Khim. Anal.*, 1953, **23**, 176.
45. P. I. Protsenko and N. P. Popovskaya, *Zh. Obshch. Khim.*, 1954, **24**, 2119.
46. P. I. Protsenko and N. P. Popovskaya, *Zh. Fiz. Khim.*, 1954, **28**, 299.
47. K. Sakai, *J. Chem. Soc. Japan, Pure Chem. Sect.*, 1954, **75**, 186.
48. N. P. Luzhnaya and I. P. Vereshchetina, *Izvest. Sekt. Fiziko-Khim. Anal.*, 1954, **24**, 192.
49. I. P. Vereshchetina and N. P. Luzhnava, *ibid.*, 1954, **25**, 188.
50. L. Shartsis and H. F. Shermer, *J. Am. Ceram. Soc.*, 1954, **37**, 544.
51. V. D. Polyakov, *Izvest. Sekt. Fiziko-Khim. Anal.*, 1955, **26**, 147.
52. P. I. Protsenko, *ibid.*, 1955, **26**, 173.
53. V. D. Polyakov, *ibid.*, 1955, **26**, 191.
54. K. B. Morris, M. I. Cook, C. Z. Sykes and M. B. Templeman, *J. Am. Chem. Soc.*, 1955, **77**, 851.
55. T. Baak, *Acta. Chem. Scand.*, 1955, **9**, 1406.
56. K. Sakai and S. Hayashi, *J. Chem. Soc. Japan, Pure Chem. Sect.*, 1955, **76**, 101.
57. E. R. van Artsdalen and I. S. Yaffe, *J. Phys. Chem.*, 1955, **59**, 118.
58. Yu. K. Delimarskii, I. N. Sheiko and V. G. Fenchenko, *Zh. Fiz. Khim.*, 1955, **29**, 1499.
59. B. S. Harrap and E. Heymann, *Trans. Faraday Soc.*, 1955, **51**, 259.
60. G. M. Pound, G. Derge and G. Osuch, *Trans. Amer. Inst. Min. Met. Eng.*, 1955, **203**, 481.
61. H. C. Cowen and H. J. Axon, *ibid.*, 1956, **52**, 242.
62. N. P. Luzhnaya, N. N. Evseeva and I. P. Vereshchetina, *Zh. Neorg. Khim.*, 1956, **1**, 1490.
63. K. Mori, *Tetsu to Hagane*, 1956, **42**, 1024.
64. B. M. Lepinskikh, O. A. Esin and S. V. Sharrin, *Zh. Priklad. Khin.*, 1956, **29**, 1813.
65. E. W. Yim and M. Feinleib, *J. Electrochem. Soc.*, 1957, **104**, 626.
66. F. R. Duke and R. A. Fleming, *ibid.*, 1957, **104**, 251.
67. J. B. Story and J. T. Clarke, *Trans. Amer. Inst. Min. Met. Eng.*, 1957, **209**, 1449.
68. V. A. Kochinashvili and V. P. Barzakovskii, *Zh. Priklad. Khim.*, 1957, **30**, 1755.
69. W. C. Phelps and R. E. Grace, *Trans. Amer. Inst. Min. Met. Eng.*, 1957, **209**, 1447.
70. S. Okado, S. Yashizawa, N. Watanabe and Y. Omota, *J. Chem. Soc. Japan, Ind. Chem. Sect.*, 1957, **60**, 670.
71. F. R. Duke and R. A. Fleming, *J. Electrochem. Soc.*, 1958, **105**, 412.
72. E. Vatslavik and A. I. Belyaev, *Zh. Neorg. Khim.*, 1958, **3**, 1044.
73. H. R. Bronstein and M. A. Bredig, *J. Am. Chem. Soc.*, 1958, **80**, 2077.
74. O. A. Esin and V. L. Zyazev, *Izvest. Akad. Nauk, S.S.S.R.*, 1958, No. 6, 7.
75. K. A. Kostanyan, *Izvest. Akad. Nauk. Armyan. S.S.S.R.*, 1958, **11**, 65.
76. R. H. Moss, *Univ. Microfilms (Ann. Arbor, Mich.)*, 1955, No. 12, 730.
77. Y. Doucet and M. Bizouard, *Compt. Rend.*, 1960, **250**, 73.
78. B. F. Markov and V. D. Prusyazhnyi, *Ukr. Khim. Zh.*, 1962, **28**, 653.
79. Y. Doucet and M. Bizouard, *Compt. Rend.*, 1959, **248**, 1328.
80. V. P. Mashovets and V. I. Petrov, *Zh. Prikl. Khim.*, 1959, **32**, 1528.
81. J. O'M. Bockris, J. A. Kitchener, S. Ignatowitz and J. W. Tomlinson, *Discuss. Faraday Soc.*, 1948, **4**, 265.
82. P. I. Protsenko and O. N. Shokina, *Zh. Neorg. Khim.*, 1960, **5**, 437.
83. P. I. Protensko and A. Ya Malakhova, *ibid.*, 1960, **5**, 2307.
84. L. F. Grantham and S. J. Yosim, *J. Chem. Phys.*, 1963, **38**, 1671.
85. A. S. Dworkin, H. R. Bronstein and M. A. Bredig, *Discuss. Faraday Soc.*, 1961, **32**, 188.
86. A. S. Dworkin, R. A. Sallach, H. R. Bronstein, M. A. Bredig and J. D. Corbett, *J. Phys. Chem.*, 1963, **67**, 1145.
87. B. F. Markov and V. D. Prusyazhnyi, *Ukr. Khim. Zh.*, 1962, **28**, 268.
88. R. V. Chernov and Yu K. Delimarskii, *Zh. Neorg. Khim.*, 1961, **6**, 2749.
89. A. S. Dworkin, H. R. Bronstein and M. A. Bredig, *J. Phys. Chem.*, 1963, **67**, 2715.
90. H. R. Bronstein and M. A. Bredig, *ibid.*, 1961, **65**, 1220.
91. H. R. Bronstein, A. S. Dworkin and M. A. Bredig, *J. Chem. Phys.*, 1961, **34**, 1843.
92. V. G. Selivanov and V. V. Stender, *Zh. Neorg. Khim.*, 1959, **4**, 2058.
93. B. F. Markov and V. D. Prusyazhnyi, *Ukr. Khim. Zh.*, 1962, **28**, 130.
94. B. F. Markov and V. D. Prusyazhnyi, *ibid.*, 1962, **28**, 419.
95. H. R. Bronstein, A. S. Dworkin and M. A. Bredig, *J. Phys. Chem.*, 1962, **66**, 44.
96. G. W. Mellors and S. Senderoff, *ibid.*, 1960, **64**, 294.
97. E. R. Buckle and P. E. Tsaoussoglou, *Trans. Faraday Soc.*, 1964, **60**, 2144.
98. M. F. Lantratov and O. F. Moiseeva, *Zh. Fiz. Khim.*, 1960, **34**, 367.

99. V. V. Rafal'skii, *Ukr. Khim. Zh.*, 1960, **26**, 585.
100. K. B. Morris and P. L. Robinson, *J. Phys. Chem.*, 1964, **68**, 1194.
101. P. C. Papaioannou and G. W. Harrington, *ibid.*, 1964, **68**, 2424.
102. H. Winterhager and L. Werner, *ForschBer. Wirt.-u. Verk Minist. N Rhein.-Westf.*, 1957, **438.**
103. K. B. Morris and P. L. Robinson, *J. Chem. Engng Data*, 1964, **9**, 444.
104. L. J. Howell and H. H. Kellogg, *Trans. Am. Inst. Min. Engrs*, 1959, **215**, 143.
105. J. O'M. Bockris and G. W. Mellors, *J. Phys. Chem.*, 1956, **60**, 1321.
106. Yu. U. Samson, L. P. Ruzinov, N. S. Rezhemnukova and V. E. Baru, *Zh. Fiz. Khim.*, 1964, **38**, 481.

Table 9.6 SURFACE TENSION OF PURE MOLTEN SALTS

The surface tension (mNm^{-1}) at temperature t (°C) is given as γ_t, or the constants a, b and t_0 in the equatior $\gamma_t = a - b(t - t_0)$ are given for the temperature range r. Principal references are in **bold type**.

AgBr	γ_{500}	152
Ref. **16, 1**	γ_{600}	148
	γ_{700}	146
AgCl	γ_{500}	176
Ref. **16, 5**	γ_{600}	171
	γ_{700}	166
AgI	γ_{600}	115
Ref. **16**		
$AgNO_3$	a	152.1
Ref. **22**, 27, 19	b	0.082
	t_0	212
	r	244–400
Al_2O_3	$\gamma_{2\,050}$	700
Ref. **28**		
B_2O_3	γ_{700}	67
Ref. **21**	$\gamma_{1\,000}$	83
	$\gamma_{1\,300}$	99
$BaBr_2$	γ_{900}	150
Ref. **20**	γ_{950}	147
	$\gamma_{1\,000}$	143
$BaCl_2$	γ_{966}	169.2
Ref. **23**, 15, 12, 9, 2	γ_{979}	168.3
	γ_{984}	168.6
	γ_{996}	168.1
	$\gamma_{1\,000}$	167.8
	$\gamma_{1\,050}$	164
$Ba(NO_3)_2$	a	134.8
Ref. **22**	b	0.015
	t_0	595
	r	600–660
$BiBr_3$	γ_{250}	66
Ref. **6**	γ_{350}	56
	γ_{450}	45
$BiCl_3$	γ_{275}	67
Ref. **6**	γ_{325}	59
	γ_{375}	53
BrF_3	γ_{12}	37
Ref. **18**	γ_{27}	36
	γ_{45}	34
BrF_5	γ_9	24
Ref. **18**	γ_{33}	22
$CaBr_2$	γ_{780}	117.31
Ref. **20**	γ_{800}	116.39
$CaCl_2$	γ_{800}	150.57
Ref. **28, 29**	$\gamma_{1\,000}$	140.07
CaI_2	γ_{800}	85
Ref. **20**	γ_{900}	83
	$\gamma_{1\,000}$	81
$Ca(NO_3)_2$	γ_{560}	101.5
Ref. **22**		
$CdBr_2$	γ_{600}	67
Ref. **16**	γ_{700}	65
$CdCl_2$	γ_{600}	83
Ref. **16**	γ_{700}	81
	γ_{800}	78
CsBr	γ_{650}	82.5
Ref. **28, 29**	γ_{800}	73.35
	$\gamma_{1\,000}$	61.147
CsCl	γ_{700}	87.50
Ref. **29, 6**, 12	γ_{900}	71.18
	$\gamma_{1\,100}$	54.86
CsF	γ_{700}	106.1
Ref. **28, 29**	γ_{900}	92.26
	$\gamma_{1\,100}$	78.46
CsI	γ_{650}	73.47
Ref. **28**, 29	γ_{800}	64.77
	$\gamma_{1\,000}$	53.17
$CsNO_3$	a	92.5
Ref. **22**, 6	b	0.069
	t_0	414
	r	421–597
Cs_2SO_4	$\gamma_{1\,000}$	114.25
Ref. **30**	$\gamma_{1\,250}$	98.84
	$\gamma_{1\,500}$	83.75
FeO	$\gamma_{1\,400}$	584.0
Ref. **28**		
$GaCl_2$	a	56.6
Ref. **24**	b	0.18
	t_0	170
	r	166–170
$GaCl_3$	γ_{80}	27
Ref. **17**	γ_{110}	24
	γ_{140}	21
GeO_2	$\gamma_{1\,000}$	241
Ref. **21**	$\gamma_{1\,200}$	253
	$\gamma_{1\,400}$	265
HgBr2	γ_{241}	64
Ref. **8**	γ_{276}	60
$HgCl_2$	γ_{293}	56
Ref. **8**		
IF_5	γ_{18}	31
Ref. **18**	γ_{28}	28
KBr	γ_{750}	90
Ref. **16, 11, 6**, 27	γ_{800}	86
	γ_{850}	82
	γ_{900}	78

(*continued*)

Table 9.6 SURFACE TENSION OF PURE MOLTEN SALTS—*continued*

KCl	γ_{800}	98
Ref. **19, 16, 15, 12, 9, 6**, 27	γ_{900}	90
	$\gamma_{1\,000}$	82
	$\gamma_{1\,100}$	75
$K_2Cr_2O_7$	γ_{420}	140.28
Ref. **30**	γ_{470}	138.28
	γ_{520}	136.28
KF	γ_{900}	138.12
Ref. **28, 29**, 30	$\gamma_{1\,100}$	123.12
	$\gamma_{1\,300}$	108.12
KI	a	138.7
Ref. **27**	b	0.087
	t_0	0
	r	(m.p. + 10)–(m.p. + 210)
K_2MoO_4	γ_{900}	151.47
Ref. **30**	$\gamma_{1\,200}$	132.87
	$\gamma_{1\,500}$	114.27
KNO_2	a	107.6
Ref. **22**	b	0.080
	t_0	435
	r	445–501
KNO_3	γ_{350}	108.61
Ref. **27**	γ_{450}	102.21
	γ_{600}	92.61
	γ_{800}	79.81
KPO_3	γ_{900}	193.54
Ref. **30**	$\gamma_{1\,200}$	171.94
	$\gamma_{1\,500}$	150.34
K_2SO_4	$\gamma_{1\,100}$	143
Ref. **13**	$\gamma_{1\,300}$	130
	$\gamma_{1\,500}$	116
K_2WO_4	γ_{900}	162.41
Ref. 30	$\gamma_{1\,200}$	134.51
	$\gamma_{1\,500}$	106.61
La_2O_3	$\gamma_{2\,320}$	560
Ref. **7**		
LiCl	γ_{600}	137.14
Ref. **28, 29**	γ_{800}	123.22
	$\gamma_{1\,000}$	109.30
LiF	γ_{850}	250.46
Ref. **28, 29**	$\gamma_{1\,050}$	228.60
	$\gamma_{1\,250}$	206.74
$LiNO_2$	a	115.4
Ref. **22**, 6	b	0.053
	t_0	255
	r	276–425
Li_2SO_4	γ_{900}	223
Ref. **13, 6**	$\gamma_{1\,000}$	216
	$\gamma_{1\,100}$	209
	$\gamma_{1\,200}$	202
NH_4NO_3	a	101.9
Ref. **22**	b	0.105
	t_0	170
	r	170–220
Na_3AlF_6	$\gamma_{1\,000}$	134.06
Ref. 31	$\gamma_{1\,010}$	132.78
	$\gamma_{1\,020}$	131.50
NaBr	γ_{750}	106
Ref. **16, 6**, 27	γ_{800}	103
	γ_{950}	92
	$\gamma_{1\,150}$	79
NaCl	γ_{800}	116.42
Ref. **6**, 27	γ_{900}	107.12
	$\gamma_{1\,000}$	97.82
	$\gamma_{1\,100}$	88.52
NaF	$\gamma_{1\,000}$	185.21
Ref. **6**, 31	$\gamma_{1\,200}$	168.81
	$\gamma_{1\,400}$	152.41
	$\gamma_{1\,500}$	144.21
NaI	a	147.4
Ref. **27**	b	0.090
	t_0	0
	r	(m.p. + 10)–(m.p. + 210)
Na_2MoO_4	γ_{700}	211.68
Ref. 30	γ_{900}	196.28
	$\gamma_{1\,100}$	180.88
	$\gamma_{1\,200}$	173.18
$NaNO_2$	a	121.2
Ref. **22**, 27	b	0.041
	t_0	277
	r	291–384
$NaNO_3$	γ_{310}	116.21
Ref. 27	γ_{400}	112.70
	γ_{500}	108.80
	γ_{600}	104.90
$NaPO_3$	γ_{800}	200.74
Ref. **6**, 30	$\gamma_{1\,000}$	186.34
	$\gamma_{1\,200}$	171.94
	$\gamma_{1\,500}$	150.34
Na_2SO_4	γ_{900}	193
Ref. **13, 6**	γ_{950}	190
	$\gamma_{1\,000}$	186
	$\gamma_{1\,050}$	183
Na_2WO_4	γ_{700}	201.46
Ref. **6**, 30	$\gamma_{1\,000}$	182.0
	$\gamma_{1\,300}$	162.46
	$\gamma_{1\,600}$	143.0
$PbCl_2$	γ_{500}	137.12
Ref. 27	γ_{600}	126.12
	γ_{700}	115.12
P_2O_5	γ_{100}	60
Ref. **21**	γ_{200}	58
	γ_{400}	54
RbBr	γ_{700}	99.77
Ref. **28, 29**	γ_{900}	88.17
	$\gamma_{1\,100}$	76.57
RbCl	γ_{750}	94.90
Ref. **28, 29**	γ_{900}	82.49
	$\gamma_{1\,000}$	74.22
	$\gamma_{1\,150}$	61.82
RbF	γ_{800}	163.83
Ref. 28, 29	γ_{900}	153.60
	$\gamma_{1\,000}$	143.37
RbI	γ_{700}	76.86
Ref. **28, 29**	γ_{900}	63.18
	$\gamma_{1\,000}$	56.34
$RbNO_3$	γ_{300}	109.53
Ref. **32**	γ_{500}	95.53
	γ_{700}	81.53
Rb_2SO_4	$\gamma_{1\,100}$	130.8
Ref. **30**	$\gamma_{1\,300}$	120.4
	$\gamma_{1\,500}$	110

(*continued*)

Table 9.6 SURFACE TENSION OF PURE MOLTEN SALTS—*continued*

SiO_2	$\gamma_{1\,000}$	278
Ref. **21**	$\gamma_{1\,500}$	295
	$\gamma_{2\,000}$	313
$SnCl_2$	γ_{300}	96.74
Ref. **28, 29**	γ_{400}	88.44
	γ_{500}	80.14
$SrBr_2$	γ_{700}	147
Ref. **20**	γ_{800}	143
	γ_{900}	138
	$\gamma_{1\,000}$	134
$SrCl_2$	γ_{900}	168
Ref. **20**	γ_{950}	165
	$\gamma_{1\,000}$	162
SrI_2	γ_{600}	112
Ref. **20**	γ_{750}	106
	γ_{850}	102
	γ_{950}	98
$Sr(NO_3)_2$	γ_{615}	128.4
Ref. **22**		
$TlNO_3$	a	94.8
Ref. **22**, 6	b	0.078
	t_0	206
	r	226–458
Tl_2S	a	215.6
Ref. **25**	b	0.035 6
	t_0	445
	r	500–700
V_2O_5	$\gamma_{1\,000}$	86
Ref. **26**		

REFERENCES TO TABLE 9.6

1. A. Gradenwitz, *Ann. Physik.*, 1899, **67**, 467.
2. Z. Motylewski, *Z. Anorg. Chem.*, 1904, **38**, 410.
3. R. Lorenz and F. Kaufler, *Ber. Dt. Chem. Ges.*, 1908, **41**, 3727.
4. R. Lorenz and A. Liebmann, *Z. Phys. Chem.*, 1913, **83**, 459.
5. R. Lorenz, A. Liebmann and A. Hochberg, *Z. Anorg. Chem.*, 1916, **94**, 301.
6. F. M. Jaeger, *Z. Anorg. Chem.*, 1917, **101**, 1.
7. H. V. Wartenberg, G. Wehner and E. Suran, *Nachr. Ges. Wiss. Göttingen*, 1936, **2**, 65.
8. E. B. R. Prideaux and J. R. Jarrett, *J. Chem. Soc.*, 1938, 1203.
9. V. P. Barzakovskii, *Bull. Acad. Sci. URSS. Classe Sci. Chim.*, 1940, 825.
10. P. P. Kozakevich and A. F. Kononenko, *J. Phys. Chem. (USSR)*, 1940, **14**, 1118.
11. K. Semenchenko and L. P. Shikhobolova, *ibid.*, 1947, **21**, 613.
12. K. Semenchenko, *ibid.*, 1947, **21**, 707.
13. K. Semenchenko, *ibid.*, 1947, **21**, 1387.
14. A. Vajna, *Alluminio*, 1951, **20**, 29.
15. J. S. Peake and M. R. Bothwell, *J. Am. Chem. Soc.*, 1954, **76**, 2625.
16. N. K. Boardman, A. R. Palmer and E. Heymann, *Trans. Faraday Soc.*, 1955, **51**, 277.
17. N. N. Greenwood and K. Wade, *J. Inorg. Nuclear Chem.*, 1957, **3**, 349.
18. M. T. Rogers and E. E. Carver. *J. Phys. Chem.*, 1958, **62**, 952.
19. J. L. Dahl and F. R. Duke, *US Atomic Energy Comm.*, 1958, ISC-923.
20. R. B. Ellis, J. E. Smith and E. B. Baker, *J. Phys. Chem.*, 1958, **62**, 766.
21. W. D. Kingery, *J. Am. Ceram. Soc.*, 1959, **42**, 6.
22. C. C. Addison and J. M. Coldrey, *J. Chem. Soc.*, 1961, 468.
23. I. D. Sokolova and N. K. Voskresenskaya, *Zh. Prikl. Khim.*, 1962, **36**, 955.
24. N. N. Greenwood and I. J. Worrall, *J. Chem. Soc.*, 1958, 1680.
25. V. B. Lazarev and M. N. Abdusalyamova, *Izv. Akad. Nauk SSSR, Ser. Khim.*, 1964, 1104.
26. B. M. Lepinskikh, O. A. Esin and G. A. Teterin, *Zh. Neorg. Khim.*, 1960, **5**, 642.
27. H. Bloom, F. G. Davis and D. W. James, *Trans. Faraday Soc.*, 1960, **56**, 1179.
28. O. K. Sokolov, *Izv. Akad. Nauk SSSR Met. Gorn. Delv*, 1963, **4**, 59.
29. R. B. Ellis and W. S. Wilcox, Work performed under U. S. At. Energy Comm; T-10-7622, 1962, pp. 128–36.
30. 'International Critical Tables', McGraw-Hill, New York, 1933.
31. H. Bloom and B. W. Burrows, 'Proc 1st Australian Conf. Electrochem' (J. A. Friend and F. Gutman eds), p. 882. Pergamon Press, Oxford, 1964.
32. S. D. Gromakov and A. I. Kostromin, *Univ. in V. I U P Yanova-Lenina Khim.*, 1955, **115**, 93.

Table 9.7 SURFACE TENSION OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS

The surface tension (mN m^{-1}) at temperature t (°C) and composition p (wt.%) of the first-named constituent is given as γ_t (or σ_t), or the constants a, b and t_0 in the equation $\gamma_t = a - b(t - t_0)$ are given for the temperature range r. Principal references are in **bold type**.

AgBr–AgCl	p	0	24.7	46.7	66.4	84.0	100	
Ref.**16**	σ_{500}	176	169	164	158	155	152	
	σ_{600}	171	164	160	155	152	148	
AgBr–AgI	p	0	16.7	34.8	54.5	76.2	100	
Ref. **16**	σ_{500}	—	122	127	133	141	152	
	σ_{600}	115	118	123	130	138	148	
AgBr–KBr	p	0	28.3	51.3	70.3	86.3	100	
Ref. **16**	σ_{700}	—	96	99	104	113	146	
	σ_{750}	90	92	96	101	110	146	
AgCl–KCl	p		32.6	56.3	72.6	88.6	100	
Ref. **16**	σ_{600}		112	117	123	136	171	
	σ_{700}		105	110	117	129	166	
AgCl–$PbCl_2$	p	0	11.4	25.6	43.7	67.4	100	
Ref. **16**	σ_{500}	137	139	144	150	160	176	
	σ_{600}	127	129	134	142	152	171	
$AgNO_3$–$CsNO_3$	p	22.5	46.5	72.4	88.8			
Ref. **19**	a	126.1	130.3	140.1	148.4			
	b	0.075	0.072	0.073	0.072			
	t_0	0	0	0	0			
	r		liquidus–c.	400				
$AgNO_3$–KNO_3	p	20	40	60	80	95	100	
Ref. **18**, 24. 19	σ_{250}	—	—	124	129	139	147	
	σ_{300}	—	118	121	126	136	144	
	σ_{350}	113	114	117	122	133	140	
$AgNO_3$–$LiNO_3$	p	45.0	71.2	88.1	100			
Ref. **19**	a	137.8	145.7	153.5	163.7			
	b	0.064	0.068	0.067	0.066			
	t_0	0	0	0	0			
	r		liquidus–c.	400				
$AgNO_3$–$NaNO_3$	p	24.1	33.3	60.0	75.0	90.0	95.0	
Ref. **18**, 24, 19	σ_{300}	—	—	126	130	136	137	
	σ_{350}	119	120	123	127	133	135	
	σ_{400}	115	116	—	—	—	—	
$AgNO_3$–$RbNO_3$	p	27.8	53.5	77.7	91.3			
Ref. **19**	a	134.0	136.9	143.4	151.7			
	b	0.077	0.073	0.070	0.069			
	t_0	0	0	0	0			
	r		liquidus–c.	400				
AlF_3–Na_3AlF_6	p	2.5	5	10	15			
Ref. **11, 1**	$\sigma_{1\,000}$	142	137	127	118.			
Al_2O_3–FeO	p	5.6	6.9					
Ref. **4**	$\sigma_{1\,410}$	598	604					
Al_2O_3–Na_3AlF_6	p	2.5	5	10	15			
Ref. **11, 1**	$\sigma_{1\,000}$	143	140	136	134			
B_2O_3–BaO	p	20.5	34.6	44.4	54.7	64.2	75.1	84.5
Ref. **14**	σ_{900}	—	—	239	197	146	88	79
	$\sigma_{1\,100}$	315	257	229	187	138	92	87
	$\sigma_{1\,300}$	294	243	221	181	135	106	96
B_2O_3–CaO	p	51.2	62.8	73.3	83.1	93.7	98.1	
Ref. **14**	σ_{900}	—	—	176	—	81	81	
	$\sigma_{1\,100}$	302	235	150	87	87	88	
	$\sigma_{1\,300}$	297	221	140	96	95	96	
B_2O_3–K_2O	p	51.5	55.7	65.6	75.3	84.9	94.8	98.5
Ref.**13**	σ_{700}	—	—	—	140	107	80	75
	σ_{900}	139	153	160	143	112	85	80
	$\sigma_{1\,100}$	—	—	149	142	118	93	88

(*continued*)

Table 9.7 SURFACE TENSION OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

B_2O_3–Li_2O	p	71.3	79.9	87.1	92.0	95.5	100	
Ref. **13**	σ_{700}	—	221	—	120	84	74	
	σ_{900}	231	207	158	121	92	80	
	$\sigma_{1\,100}$	—	—	152	125	101	88	
B_2O_3–Na_2O	p	64.0	73.6	82.2	85.8	94.4	99.1	
Ref. **13**	σ_{700}	—	196	146	—	83	77	
	σ_{900}	197	185	147	130	91	81	
	$\sigma_{1\,100}$	—	—	149	134	99	89	
B_2O_3–PbO	p	0	15.9	33.6	50.9	65.6	78.4	100
Ref. **10, 8**	σ_{700}	—	162	160	—	—	—	—
	σ_{900}	132	163	144	92	78	79	80
	$\sigma_{1\,000}$	135	163	—	—	—	—	83
B_2O_3–SrO	p	35.4	49.6	61.6	70.0	89.0	96.0	
Ref. **14**	σ_{900}	—	—	194	137	80	—	
	$\sigma_{1\,100}$	303	247	178	123	87	86	
	$\sigma_{1\,300}$	292	237	166	122	95	93	
B_2O_3–ZnO	p	20.5	29.4	35.4	41.4	50.0	90.0	100
Ref. **9**	σ_{900}	—	—	235	184	80	78	80
	$\sigma_{1\,100}$	—	283	232	179	88	86	86
	$\sigma_{1\,300}$	355	287	238	185	—	95	94
$BaCl_2$–KCl	p	0	20.4	47.4	59.0	67.2	82.2	92.8
Ref. **15**	σ_{850}	93	98	106	110	114	129	—
	σ_{900}	89	94	103	106	112	128	144
	σ_{950}	—	—	—	—	—	121	141
$BaCl_2$–Li_2SO_4	p	0	9.0	17.3	44.6	65.3	84.9	100
Ref. **6**	$\sigma_{1\,000}$	220	198	192	167	163	161	175
	$\sigma_{1\,050}$	216	190	180	164	158	159	172
$BaCl_2$–NaCl	p	0	28.4	47.1	65.7	78.1	93.4	100
Ref. **3**, 20	σ_{900}	107	109	117	126	133	160	—
	$\sigma_{1\,000}$	100	103	111	120	126	154	171
$CaCl_2$–NaCl*	p	0	17.4	44.8	65.5	88.3	100	
Ref. **3**	σ_{800}	114	115	121	126	137	148	
	$\sigma_{1\,000}$	100	100	108	116	125	140	
CaF_2–Na_3AlF_6	p	2.5	5	10	15			
Ref. **11**	$\sigma_{1\,000}$	149	150	152	155			
CaO–FeO	p	0	5.5	15.4	22.3			
Ref. **4**	$\sigma_{1\,410}$	585	555	543	573			
$CdBr_2$–$CdCl_2$	p	0	20.9	41.3	61.7	80.9	100	
Ref. **16**	σ_{600}	83	78	74	71	68	67	
	σ_{700}	81	76	72	68	66	65	
$CdCl_2$–KCl	p		38.1	62.1	78.7	90.7	100	
Ref. **16**	σ_{600}		100	93	90	88	82	
	σ_{700}		94	87	84	84	81	
$CdCl_2$–NaCl	p	0	44.0	67.6	82.5	92.6	100	
Ref. **16**	σ_{700}	—	108	98	90	85	81	
	σ_{800}	114	101	92	85	81	78	
$CdCl_2$–$PbCl_2$	p	0	14.1	31.8	49.7	72.4	100	
Ref. **16**	σ_{600}	126	116	107	98	90	83	
	σ_{700}	116	108	101	93	87	81	
Cr_2O_3–FeO	p	0	3.1					
Ref. **4**	$\sigma_{1\,420}$	585	588					
CsCl–Li_2SO_4	p	0	1.6	3.2	4.8	15.2	61.8	100
Ref. **7, 6**	σ_{900}	224	205	193	189	163	102	72
	$\sigma_{1\,000}$	220	201	188	185	160	92	64
	$\sigma_{1\,100}$	211	194	181	177	154	—	—
CsCl–$PbCl_2$	p	12.4	27.4	38.6	49.0	64.7		
Ref. **18**	σ_{500}	116	—	—	—	103		
	σ_{600}	—	97	—	93	94		
	σ_{625}	—	94	91	90	92		

(*continued*)

Table 9.7 SURFACE TENSION OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$CsNO_3$–KNO_3	p	39.1	66.0	85.3				
Ref. **19**	a	133.7	129.4	125.1				
	b	0.079	0.077	0.074				
	t_0	0	0	0				
	r		liquidus–c.	400				
$CsNO_3$–$LiNO_3$	p	48.5	73.9	89.4				
Ref. **19**	a	124.3	124.3	122.9				
	b	0.070	0.076	0.075				
	t_0	0	0	0				
	r		liquidus–c.	400				
$CsNO_3$–$NaNO_3$	p	43.1	67.6	87.3				
Ref. **19**	a	130.5	127.3	123.4				
	b	0.068	0.074	0.072				
	t_0	0	0	0				
	r		liquidus–c.	400				
Cs_2O–SiO_2	p	47.2	54.0	69.0	78.9			
Ref. **21**	$\gamma_{1\,300}$	166.1	165.1	144.3	120.5			
	$\gamma_{1\,400}$	163.8	162.5	—	—			
FeO–MnO	p	90.5	94.0	100				
Ref. **4**	$\sigma_{1\,410}$	555	567	585				
FeO–SiO_2	p	68.1	76.9	84.1	95.8	100		
Ref. **4**	$\sigma_{1\,410}$	409	468	503	563	585		
FeO–TiO_2	p	81.9	85.1	100				
Ref. **4**	$\sigma_{1\,410}$	510	522	585				
KBr–KCl	p	0	28.6	51.7	70.7	86.5	100	
Ref. **16**	σ_{750}	—	97	95	93	92	90	
	σ_{800}	99	95	93	89	87	86	
KBr–NaBr	p	0	22.4	43.5	63.4	82.2	100	
Ref. **16**	σ_{700}	—	103	98	96	95	—	
	σ_{800}	103	96	92	88	87	86	
KBr–Na_2SO_4	p	0	4.2	21.8	45.6	100		
Ref. **5**	σ_{900}	193	181	140	116	81		
KCl–K_2SO_4	p	0	0.9	2.2	4.5			
Ref. **7**	$\sigma_{1\,075}$	144	140	135	133			
KCl–Li_2SO_4	p	0	7.0	26.6	40.2	66.8	85.8	100
Ref. **7, 6**	σ_{900}	224	182	145	128	109	97	91
	$\sigma_{1\,000}$	220	178	142	123	101	90	85
	$\sigma_{1\,100}$	211	173	135	116	92	81	75
KCl–$MgCl_2$	p	8.0	16.4	34.4	54.1	75.9	87.6	
Ref. **17**	σ_{700}	76	79	83	85	93	97	
	σ_{800}	69	72	76	78	86	90	
	σ_{900}	62	65	69	71	79	83	
KCl–NaCl*	p	0	24.2	46.0	65.6	83.6	100	
Ref. **18, 17, 3**	σ_{700}	—	—	114	111	—	—	
	σ_{750}	—	115	111	107	104	—	
	σ_{800}	114	111	107	104	101	96	
KCl–NaI	p	7.9	19.5	32.6	49.0	73.2	84.9	
Ref. **24**	a	135.2	128.0	121.7	124.4	127.2	128.5	
	b	0.066	0.063	0.057	0.063	0.069	0.070	
	t_0	0	0	0	0	0	0	
	r		(m.p. + 10) − (m.p. + 210)					
KCl–Na_2SO_4	p	0	2.7	14.8	34.2	100		
Ref. **5**	σ_{900}	193	183	153	129	92		
KCl–$PbCl_2$	p	0	5.5	15.2	28.6	51.7		
Ref. **18, 16**	σ_{500}	137	125	118	117	118		
	σ_{600}	117	114	109	107	107		
KI–Na_2SO_4	p	0	5.8	11.5	28.0	53.9	100	
Ref. **5**	σ_{900}	193	177	157	127	100	73	

(*continued*)

Table 9.7 SURFACE TENSION OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

KNO_2–KNO_3	p	17.4	35.8	55.9	77.0			
Ref. **24**	a	129.6	131.3	130.6	131.8			
	b	0.059	0.061	0.055	0.052			
	t_0	0	0	0	0			
	r		(m.p. + 10) − (m.p. + 210)					
KNO_3–$LiNO_3$	p	0	32.9	59.5	81.5	100		
Ref. **19**	a	129.9	127.9	129.6	133.6	139.8		
	b	0.055	0.056	0.062	0.070	0.081		
	t_0	0	0	0	0	0		
	r		liquidus–c.	400				
KNO_3–$NaNO_3$*	p	0	40	60	100			
Ref. **18, 16**, 19	σ_{350}	117	115	114	112			
	σ_{400}	114	112	111	109			
	σ_{450}	111	109	107	105			
KNO_3–$RbNO_3$	p	0	18.6	40.7	67.4			
Ref. **19**	a	134.3	135.0	136.9	138.5			
	b	0.083	0.082	0.083	0.083			
	t_0	0	0	0	0			
	r		liquidus–c.	400				
K_2O–SiO_2	p	23.9	26.7	29.9	32.9	38.7	43.6	
Ref. **12**, 21	$\sigma_{1\,000}$	—	227	227	227	225	—	
	$\sigma_{1\,200}$	222	223	220	219	218	215	
	$\sigma_{1\,400}$	219	219	216	214	210	206	
K_2SO_4–$RbCl$	p	92.9	96.5	98.6	100			
Ref. **7**	$\sigma_{1\,075}$	134	139	141	143			
$LiCl$–$PbCl_2$	p	6.1	7.8	12.8	13.4	24.9	27.1	
Ref. **18**	σ_{500}	133	134	134	133	—	—	
	σ_{550}	130	129	129	129	128	—	
	σ_{600}	124	—	—	—	—	124	
$LiCl$–$RbCl$	p	0	75.9	86.9	91.9	94.5	97.2	100
Ref. 7	σ_{750}	96	113	118	119	121	122	127
Li_2O–SiO_2	p	12.9	17.8	23.9	27.8	33.5	43.1	49.1
Ref. **12**, 21	$\sigma_{1\,100}$	—	315	328	338	—	—	—
	$\sigma_{1\,300}$	311	317	328	334	352	369	381
	$\sigma_{1\,400}$	316	317	328	332	349	364	374
$LiNO_3$–$RbNO_3$	p	13.5	31.8	58.3				
Ref. **19**	a	130.0	129.6	125.0				
	b	0.075	0.074	0.059				
	t_0	0	0	0				
	r		liquidus–c.	400				
Li_2SO_4–$NaCl$*	p	0	38.8	65.5	81.6	94.5	98.7	100
Ref. **7, 6**	σ_{900}	109	131	148	168	198	208	224
	$\sigma_{1\,000}$	104	125	143	164	194	204	220
	$\sigma_{1\,100}$	95	116	134	157	187	196	211
Li_2SO_4–$RbCl$	p	0	38.0	57.9	68.1	78.6	89.2	100
Ref. **7, 6**	σ_{900}	83	105	126	140	155	179	224
	$\sigma_{1\,000}$	74	97	120	134	149	174	220
	$\sigma_{1\,100}$	66	92	114	—	145	168	211
$MgCl_2$–$NaCl$	p	17.9	33.0	56.7	74.7	88.7	94.7	
Ref. **17**	σ_{700}	110	103	95	87	79	75	
	σ_{800}	104	97	89	81	73	69	
	σ_{900}	98	91	83	75	67	63	
Na_3AlF_6–NaF	p	85	90	95	97.5			
Ref. **11, 1**	$\sigma_{1\,000}$	161	159	155	152			
$NaCl$–$NaNO_3$	p	0	9.2					
Ref. **5**	σ_{470}	110	110					
$NaCl$–Na_2SO_4	p	0	0.8	1.7	4.3			
Ref. **7**	σ_{900}	192	189	184	179			

(*continued*)

Table 9.7 SURFACE TENSION OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$NaCl–PbCl_2$	p	0	4.9	11.7	15.1	18.2		
Ref. **18, 3**, 24	σ_{500}	—	133	131	—	—		
	σ_{550}	131	127	124	—	—		
	σ_{575}	128	124	122	122	122		
$NaF–NaNO_3$	p	0	9.8					
Ref. **5**	σ_{560}	108	108					
$NaI–NaNO_3$	p	0	10.1					
Ref. **5**	σ_{312}	119	119					
$NaI–Na_2SO_4$	p	0.5	1.1	2.1	3.2	5.3	10.5	
Ref. **2**	σ_{900}	168	167	160	158	154	141	
$NaNO_2–NaNO_3$	p	16.9	35.4	55.0	76.3			
Ref. **24**	a	127.9	127.1	128.9	126.8			
	b	0.039	0.035	0.038	0.030			
	t_0	0	0	0	0			
	r		(m.p. + 10)–(m.p. + 210)					
$NaNO_3–RbNO_3$	p	16.1	36.5	63.4				
Ref. **19**	a	132.5	133.7	135.4				
	b	0.076	0.073	0.068				
	t_0	0	0	0				
	r	liquidus–c.	400					
$Na_2O–P_2O_5$	$\gamma_t = 150.6 + 155p/(100 - p) - 0.037\,9t$							
Ref. **22**	r_p	30.4–39.5						
	r_t	liquidus–1050						
$Na_2O–SiO_2$	p	20.0	30.8	33.6	36.9	50.0		
Ref. **12, 2**, 21	$\sigma_{1\,000}$	277	284	286	288	—		
	$\sigma_{1\,200}$	276	280	281	283	295		
	$\sigma_{1\,400}$	273	274	274	276	284		
Na_2SO_4RbCl	p	0	1.7	3.4	4.3	8.6		
Ref. **7**	$\sigma_{1\,050}$	183	173	171	169	162		
$PbCl_2–RbCl$	p	46.7	62.8	69.0	83.1	90.3	98.2	
Ref. **7**	σ_{475}	—	112	112	116	124	—	
	σ_{525}	—	107	107	111	118	130	
	σ_{575}	104	104	104	107	113	124	
$PbO–SiO_2$	p	65.1	69.9	75.7	82.5	84.7	90.6	96.8
Ref. **8**, 23	σ_{900}	—	—	217	199	192	174	134
	$\sigma_{1\,100}$	234	232	221	204	196	179	145
	$\sigma_{1\,300}$	235	230	223	209	—	183	158
$PbO–V_2O_5$	p	15.3	39.6	52.8	65.4	79.0	85.1	92.1
Ref. **23**	$\gamma_{1\,000}$	92	135	168	174	205	202	192
$PbO–SiO_2–V_2O_5$	p_{PbO}	28.5	38.6	58.4	60.0	68.5	80.0	
Ref. **23**	p_{SiO_2}	10.3	10.4	10.4	10.1	10.6	10.0	
	$\gamma_{1\,000}$	128	142	150	190	210	202	
$Rb_2O–SiO_2$	p	39.1	43.9	49.5	59.5	67.3		
Ref. **21**	$\gamma_{1\,200}$	—	—	—	175.1	155.0		
	$\gamma_{1\,300}$	200.1	192.7	188.0	173.4	146.3		
	$\gamma_{1\,400}$	197.1	188.8	183.5	170.9	—		

* *See also:* 'Physical Properties Data Compilations Relevant to Energy Storage II Molten Salts: Data on Single and Multi-Component Salt Systems', Janz *et al.*, NSRDS–NBS 61, Part II.

REFERENCES TO TABLE 9.7

1. E. Elchardus, *Compt. Rend.*, 1938, **206**, 1460.
2. C. W. Parmelee and C. G. Harman, *Univ. Illinois Eng. Exptl. Sta. Bull.*, 1939, No. 311, 29.
3. V. P. Barzakovskii, *Bull Acad. Sci. U.R.S.S., Classe Sci. Chim.*, 1940, 825.
4. P. P. Kozakevich and A. F. Kononenko, *J. Phys. Chem. (USSR)*, 1940, **14**, 1118.
5. K. Semenchenko and L. P. Shikhobolova, *ibid.*, 1947, **21**, 613.
6. K. Semenchenko, *ibid.*, 1947, **21**, 707.
7. K. Semenchenko, *ibid.*, 1947, **21**, 1387.

8. L. Shartsis, S. Spinner and A. W. Smock, *J. Am. Ceram. Soc.*, 1948, **31**, 23.
9. L. Shartsis and R. Canga, *J. Res. Nat. Bur. Stand.*, 1949, **43**, 221.
10. S. Carlen, *Trans. Chalmers Univ. Tech. Gottenburg*, 1949, No. 85.
11. A. Vajna, *Alluminio*, 1951, **20**, 29.
12. L. Shartsis and S. Spinner, *J. Res. Nat. Bur. Stand.*, 1951, **46**, 385.
13. L. Shartsis and W. Capps, *J. Am. Ceram. Soc.*, 1952, **35**, 169.
14. L. Shartsis and H. F. Shermer, *ibid.*, 1954, **37**, 544.
15. J. S. Peake and M. R. Bothwell, *J. Am. Chem. Soc.*, 1954, **76**, 2656.
16. N. K. Boardman, A. R. Palmer and E. Heymann, *Trans. Faraday Soc.*, 1955, **51**, 277.
17. O. G. Desyatnikov, *Zh. Priklad. Khim.*, 1956, **29**, 870.
18. J. H. Dahl and F. R. Duke, *J. Phys. Chem.*, 1958, **62**, 1498.
19. G. Bertozzi and G. Sternheim, *ibid.*, 1964, **68**, 2908.
20. I. D. Sokolova and N. K. Voskresenskaya, *Zh. Prkl. Khim.*, 1962, **36**, 955.
21. A. A. Appen and S. S. Kayalova, *Dokl. Akad. Nauk., SSSR, Ser. Fiz. Khim.*, 1962, **145**, 592.
22. C. F. Callis, J. R. Van Wazer and J. S. Metcalf, *J. Am. Chem. Soc.*, 1955, **77**, 1468.
23. B. M. Lepinskikh, O. A. Esin and G. A. Teterin, *Zh. Neorg. Khim.*, 1960, **5**, 642.
24. H. Bloom, F. G. Davis and D. W. James, *Trans. Faraday Soc.*, 1960, **56**, 1179.

Table 9.8 VISCOSITY OF PURE MOLTEN SALTS

The viscosity (centipoise) at temperature t (°C) is given as η_t, or the constants a and b in the equation $\log \eta_t = a + b/(t + 273)$ are given for the temperature range r. Principal references are in **bold type**.

AgBr	η_{450}	3.3
Ref. **22, 11, 9**	η_{550}	2.4
	η_{650}	1.7
	η_{800}	1.2
AgCl	η_{500}	2.05
Ref. **22, 11, 9**	η_{600}	1.60
	η_{700}	1.25
	η_{800}	1.05
AgI	η_{600}	3.0
Ref. **11**, 9	η_{700}	2.3
	η_{800}	1.7
$AgNO_3$	η_{230}	4.10
Ref. **25, 21, 17, 11, 9**	η_{280}	3.05
	η_{330}	2.40
	η_{350}	2.20
B_2O_3	η_{600}	158 000
Ref. **24, 20, 4**	η_{800}	25 100
	$\eta_{1\,000}$	6 300
	$\eta_{1\,200}$	2 000
$BaCl_2$	$\eta_{1\,000}$	4.506
Ref. 2, **6**		
$BiCl_3$	η_{260}	32
Ref. **5**	η_{300}	23
	η_{340}	18
$CaCl_2$	η_{800}	3.021
Ref. **6**, 7	η_{900}	1.870
	$\eta_{1\,000}$	1.248
$CdBr_2$	η_{600}	2.60
Ref. **18**	η_{640}	2.35
	η_{680}	2.10
$CdCl_2$	η_{600}	2.35
Ref. **23, 18, 14**	η_{700}	1.85
	η_{800}	1.55
CuCl	η_{500}	2.80
Ref. **14**	η_{600}	1.95
	η_{700}	1.40
$HgBr_2$	η_{255}	2.196
Ref. 8, **13**	η_{265}	2.008
	η_{275}	1.843
$HgCl_2$	η_{281}	1.768
Ref. **32**	η_{287}	1.738
	η_{292}	1.694
	η_{299}	1.600
	η_{306}	1.543
HgI_2	η_{268}	2.669
Ref. **32**	η_{292}	2.244
	η_{314}	1.995
	η_{334}	1.715
	η_{358}	1.458
KBr	η_{750}	1.150
Ref. 7, **14**, 16	η_{800}	1.022
	η_{900}	0.831
KCl	η_{800}	1.094
Ref. **14**, 16, 26	η_{900}	0.841
	$\eta_{1\,000}$	0.673
$K_2Cr_2O_7$	η_{400}	13.79
Ref. **27**, 29	η_{450}	9.665
	η_{500}	7.091
KI	η_{750}	1.362
Ref. 30	η_{800}	1.205
	η_{900}	0.973
KNO_2	a	−0.87
Ref. **33**	b	960
	r	418–450
KNO_3	η_{350}	2.705
Ref. 27, 16, 31, **34**, 35	η_{400}	2.090
	η_{450}	1.673
	η_{550}	1.163
KOH	η_{400}	2.3
Ref. **10**	η_{500}	1.3
	η_{600}	0.8
LiBr	η_{550}	1.815
Ref. 7, **14**	η_{700}	1.096
	η_{850}	0.757
LiI	η_{450}	2.50
Ref. **14**	η_{550}	1.70
	η_{650}	1.30
$LiNO_3$	η_{260}	5.5
Ref. **17, 12, 11, 3**	η_{300}	4.0
	η_{350}	2.9
Na_3AlF_6	$\eta_{1\,007}$	6.7
Ref. **19**	$\eta_{1\,017}$	6.5
	$\eta_{1\,050}$	6.0
NaBr	η_{762}	1.345
Ref. **14**, 16	η_{766}	1.332
	η_{780}	1.288
NaCl	η_{800}	1.463
Ref. 16, **34**, 35	η_{900}	1.009
	$\eta_{1\,000}$	0.737
NaI	η_{650}	1.581
Ref. **30**	η_{750}	1.168
	η_{900}	0.818
$NaNO_2$	a	−1.07
Ref. **33**	b	868
	r	282–310
$NaNO_3$	η_{300}	3.156
Ref. 31, **34**, 36, 37	η_{400}	1.901
	η_{500}	1.305
NaOH	η_{350}	4.0
Ref. **10**	η_{450}	2.2
	η_{550}	1.5
$NaPO_3$	η_{650}	1 250
Ref. **28, 4**	η_{750}	440
	η_{850}	210
	η_{900}	150
$PbBr_2$	η_{360}	10.5
Ref. **22, 17, 1**	η_{400}	7.3
	η_{450}	5.0
	η_{550}	3.0
$PbCl_2$	η_{500}	4.6
Ref. **22, 1**	η_{600}	2.8
	η_{700}	1.9
$TlNO_3$	a	−1.04
Ref. **33**	b	565
	r	207–250

REFERENCES TO TABLE 9.8

1. R. Lorenz and H. T. Kalmus, *Z. Phys. Chem.*, 1907, **59**, 244.
2. V. T. Slavyanskii, *Dokl. Akad. Nauk. SSSR*, 1947, **58**, 1077.
3. H. M. Goodwin and R. D. Mailey, *Phys. Rev.*, 1908, **26**, 28.
4. K. Arndt, *Z. Chem. Apparat.*, 1908, **3**, 549.
5. A. H. W. Aten, *Z. Phys. Chem.*, 1909, **66**, 641.
6. G. J. Janz and R. D. Reeves, *Advan. Electrochem. Eng.*, 1967, **5.**
7. S. Karpachev and A. Stromberg, *Zh. Fiz. Khim.*, 1938, **11**, 852.
8. R. S. Dantuma, *Z. Anorg. Allgem. Chem.*, 1938, **175**, 1.
9. R. Lorenz and A. Hoechberg, *Z. Anorg. Chem.*, 1916, **94**, 317.
10. K. Arndt and G. Ploetz, *Z. Phys. Chem.*, 1926, **121**, 439.
11. E. van Aubel, *Bull. Sci. Acad. Roy. Belg.*, 1926, **12**, 374.
12. R. S. Dantuma, *Z. Anorg. Allg. Chem.*, 1928, **175**, 1.
13. G. Jander and K. Broderson, *Z. Anorg. Allgem. Chem.*, 1951, **264**, 57.
14. I. G. Murgulescu and S. Zuca, *Z. Physik. Chem.* (*Leipzig*), 1961, **218**, 379.
15. A. G. Stromberg, *Zh. Fiz. Khim.*, 1939, **13**, 436.
16. C. E. Fawsitt, *Proc. Roy. Soc.* (*London*), 1908, 93.
17. K. S. Evstropev, *Akad. Nauk. SSSR., Otdel. Tekh. Nauk. Inst. Mash. Sov.*, 1945, **3**, 61.
18. H. Bloom, B. S. Harrap and E. Heymann, *Proc. R. Soc.*, 1948, **A194**, 237.
19. A. Vajna, *Alluminio*, 1950, **19**, 133.
20. L. Shartsis, W. Capps and S. Spinner, *J. Am. Ceram. Soc.*, 1953, **36**, 319.
21. F. A. Pugsley and F. E. W. Wetmore, *Canad. J. Chem.*, 1954, **32**, 839.
22. B. S. Harrap and E. Heymann, *Trans. Faraday Soc.*, 1955, **51**, 259.
23. B. S. Harrap and E. Heymann, *ibid.*, 1955, **51**, 268.
24. J. D. Mackenzie, *ibid.*, 1956, **52**, 1564.
25. N. P. Luzhnaya, N. N. Evseeva and I. P. Vereshchetina, *Zh. Neorg. Khim.*, 1956, **1**, 1490.
26. S. Karpachev, *Zh. Obshch. Khim.*, 1935, **5**, 625.
27. R. Lorenz and T. Kalmus, *Z. Physik. Chem.*, 1907, **59**, 244.
28. G. G. Nozadze, *Soobshch. Akad. Nauk. Gruzin. S.S.S.R.*, 1957, **19**, 567.
29. J. P. Frame, E. Rhodes and A. R. Ubbelohde, *Trans. Faraday Soc.*, 1959, **55**, 2039.
30. I. G. Murgulescu and S. Zuca, *Rev. Roumaine. Chim.*, 1965, **10**, 123.
31. H. M. Goodwin and R. D. Mailey, *Phys. Rev.*, 1906, **23**, 22; *ibid.*, 1907, **25**, 469; *ibid.*, 1908, **26,** 28.
32. G. J. Janz and J. D. E. McIntyre, *J. Electrochem. Soc.*, 1962, **109**, 842.
33. J. P. Frame, E. Rhodes and A. R. Ubbelohde, *Trans. Faraday Soc.*, 1959, **55**, 2039.
34. R. S. Dantuma, *Z. Anorg. Allgem. Chem.*, 1938, **175**, 1.
35. K. Ogawa, *Nippon Kinzoku Gakkaishi*, 1950, **14B**, 49.
36. R. Lorenz and H. T. Kalmus, *Z. Physik. Chem.*, 1907, **59**, 17.
37. C. E. Fawsitt, *Proc. Roy. Soc.* (*London*), 1908, **A80**, 290.

Table 9.9 VISCOSITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS

The viscosity (centipoise) at temperature t (°C) and composition p (wt. %) of the first-named constituent is given as η_t, or the constants a and b in the equation $\log \eta_t = a + b/(t + 273)$ are given for the temperature range r. Principal references are in **bold type.**

AgBr–AgCl	p	0	32.5	46.7	66.9	84.0	100	
Ref. **12**	η_{440}	—	2.58	2.75	2.97	3.12	3.38	
	η_{520}	1.98	2.10	2.18	2.35	2.46	2.69	
	η_{600}	1.68	1.74	1.74	1.98	2.09	2.28	
AgBr–KBr	p	57.1	66.0	78.0	86.0	100		
Ref. **12**	η_{400}	—	—	3.55	3.55	—		
	η_{500}	—	2.55	2.40	2.46	2.83		
	η_{600}	1.73	1.74	1.78	1.86	2.27		
AgCl–KCl	p	54.3	61.1	80.4	88.9	100		
Ref. **12**	η_{500}	—	—	2.10	2.12	2.08		
	η_{600}	1.63	1.61	1.56	1.62	1.66		
	η_{700}	1.28	1.24	1.24	1.28	1.40		
AgCl–$PbCl_2$	p	0	9.8	24.1	44.5	67.6	100	
Ref. **12**	η_{500}	4.56	4.13	3.47	2.95	2.47	2.08	
	η_{600}	2.75	2.59	2.30	2.12	1.84	1.66	
	η_{700}	1.87	1.99	1.84	1.76	1.65	1.40	
AgI–$AgNO_3$	p	0	25.7	47.9	67.4			
Ref. **17**	η_{150}	—	9.6	13.6	19.6			
	η_{200}	—	6.4	8.6	12.6			
	η_{250}	4.2	5.2	6.5	8.7			
	η_{300}	3.8	4.8	5.5	7.0			
$AgNO_3$–HgI_2	p	9.8	13.8	19.9	27.2	35.9	46.6	56.2
Ref. **15, 21**	η_{110}	—	2.3	—	—	19.5	13.8	9.6
	η_{150}	3.2	—	7.4	7.0	5.6	4.2	3.2
	η_{200}	1.5	2.3	2.4	2.2	1.9	1.8	1.4
$AgNO_3$–KNO_3	p	29.6	41.9	52.8	62.7	71.6	79.7	83.5
Ref. **14, 21**	η_{150}	—	—	—	18.2	19.1	18.1	—
	η_{200}	—	—	—	8.9	9.6	9.2	8.9
	η_{250}	—	7.4	6.4	5.6	6.0	6.4	6.0
	η_{300}	5.8	5.1	4.8	4.1	4.3	4.6	4.5
AlF_3–Na_3AlF_6	p	2.5	5	10	15			
Ref. **7**	$\eta_{1\,000}$	6.5	6.3	5.7	4.8			
Al_2O_3–Na_3AlF_6	p	2.5	5	10	15			
Ref. **7**	$\eta_{1\,000}$	6.9	6.9	7.1	10.9			
B_2O_3–BaO	p	44.5	49.2	54.9	59.7	64.4	67.9	
Ref. **10**	η_{850}	—	30 900	35 500	35 500	33 900	25 100	
	η_{900}	2 750	6 170	8 710	9 550	9 550	7 080	
	η_{950}	980	1 900	3 160	3 710	3 240	2 950	
	$\eta_{1\,000}$	—	—	—	1 260	1 260	1 180	
B_2O_3–K_2O	p	61.6	70.6	84.9	89.0	94.8	98.5	100
Ref. **9**	η_{600}	6.8×10^5	5.1×10^6	—	1.5×10^5	0.9×10^5	1.3×10^5	1.6×10^5
	η_{800}	780	4 680	4 360	3 900	5 130	13 800	21 400
	η_{900}	—	960	1 350	1 290	2 400	6 310	11 500
	$\eta_{1\,000}$	—	—	—	450	1120	3710	6460
B_2O_3–Li_2O	p	85.2	87.1	92.0	95.5	97.2	100	
Ref. **9**	η_{600}	—	—	—	355 000	141 000	158 000	
	η_{800}	3 630	3 020	5 370	3 630	7 240	21 400	
	η_{900}	460	520	790	980	3 090	11 500	
	$\eta_{1\,000}$	—	—	250	320	1 440	6 460	
B_2O_3–Na_2O	p	68.2	69.2	77.6	85.8	94.4	99.1	100
Ref. **9**	η_{600}	—	3.1×10^6	—	1.9×10^6	1.3×10^5	1.3×10^5	1.6×10^5
	η_{800}	930	1 290	6 920	5 890	3 800	12 000	21 400
	η_{900}	—	—	870	1 380	1 580	5 750	11 500
	$\eta_{1\,000}$	—	—	—	—	810	2 630	6 460
B_2O_3–$NaPO_3$	p	0	50	95	99.5			
Ref. **18**	η_{900}	150	450	4 730	11 000			

(*continued*)

Table 9.9 VISCOSITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$BaCl_2$–NaCl	p	45.5	55.6	63.8	70.5	75.3	80.1	84.4
Ref. **19**, 5, **21**	η_{725}	—	3.2	3.42	4.05	4.6	—	—
	η_{775}	2.40	2.58	2.82	3.22	3.48	3.70	3.85
	η_{825}	2.00	2.23	2.50	2.80	3.00	3.15	3.28
	η_{875}	1.75	2.02	2.36	2.55	2.72	2.84	3.05
BaO–SiO_2	p	31.7	46.1	53.1	63.2	71.7		
Ref. **16**	$\eta_{1\,500}$	—	11 000	3 200	1 700	—		
	$\eta_{1\,700}$	19 500	2 140	850	540	190		
	$\eta_{1\,800}$	10 000	1 250	530	400	150		
$CaCl_2$–NaCl*	p	0	17.4	44.8	65.5	88.3	100	
Ref. **5**, 19	η_{800}	1.59	1.65	2.59	3.49	4.36	4.92	
	η_{900}	1.00	1.16	2.11	2.95	3.69	4.22	
	$\eta_{1\,000}$	0.70	1.00	1.79	2.60	3.31	3.74	
CaF_2–Na_3AlF_6	p	2.5	5	10	15			
Ref. **7**	$\eta_{1\,000}$	6.9	7.0	7.3	8.0			
CaO–SiO_2	p	29.1	37.1	41.9	47.9	51.9	55.9	
Ref. **11, 3**	$\eta_{1\,500}$	—	1440	765	—	288	—	
	$\eta_{1\,600}$	—	730	405	218	157	113	
	$\eta_{1\,700}$	1360	392	235	133	96	74	
	$\eta_{1\,800}$	850	250	150	88	66	54	
$CdBr_2$–$CdCl_2$	p	0	27.2	49.9	69.1	85.6	100	
Ref. **6**	η_{600}	2.3	2.4	2.4	2.5	2.5	2.6	
	η_{660}	2.0	2.0	2.1	2.1	2.1	2.2	
$CdCl_2$–KCl	p	40.3	59.5	67.3	71.9	82.4	88.8	100
Ref. **13**	η_{500}	—	2.53	2.54	2.62	2.97	3.17	—
	η_{600}	1.76	1.67	1.67	1.73	1.92	2.15	2.31
	η_{700}	1.23	1.21	1.20	1.24	1.40	1.57	1.83
$CdCl_2$–NaCl	p	53.9	75.8	79.0	82.5	90.2	100	
Ref. **13**	η_{500}	—	2.68	2.71	2.84	2.99	—	
	η_{600}	—	1.78	1.80	1.88	2.02	2.31	
	η_{700}	1.37	1.30	1.27	1.39	1.50	1.83	
$CdCl_2$–$PbCl_2$	p	0	18.0	26.8	37.9	65.9	100	
Ref. **13**	η_{520}	4.22	3.75	3.50	3.36	—	—	
	η_{600}	2.75	2.60	2.47	2.38	2.26	2.31	
	η_{680}	2.00	2.02	1.88	1.85	1.85	1.90	
KCl–$MgCl_2$	p	37.6	48.5	58.9	59.9	69.1	78.8	
Ref. **2, 21**	η_{500}	3.26	—	2.78	2.69	—	—	
	η_{600}	2.19	1.55	1.51	1.70	1.44	—	
	η_{700}	—	1.22	—	—	0.99	1.25	
KCl–NaCl*	p	0	56.0	79.3	100			
Ref. **5**	η_{800}	1.59	1.17	1.07	1.13			
	η_{900}	1.00	0.90	0.81	0.89			
KCl–$PbCl_2$	p	0	5.2	8.8	18.3	21.8	34.7	43.3
Ref. **13**	η_{500}	4.56	4.02	3.66	3.34	3.37	—	—
	η_{600}	2.75	2.42	2.27	2.06	2.10	2.15	—
	η_{700}	1.87	1.65	1.56	1.42	1.47	1.48	1.50
K_2O–SiO_2	p	3.9	9.5	15.9	23.9	29.9	38.7	43.6
Ref. **16, 8**	$\eta_{1\,000}$	—	—	—	32 800	15 100	7 940	4 570
	$\eta_{1\,200}$	—	—	—	2 240	1 350	661	355
	$\eta_{1\,400}$	—	—	—	372	200	81	47
	$\eta_{1\,600}$	468 000	107 000	24 000	—	—	—	—
Li_2O–SiO_2	p	12.0	16.1	21.8	25.9	28.9	33.2	37.9
Ref. **16, 8**	$\eta_{1\,100}$	—	50 100	12 000	4470	—	—	—
	$\eta_{1\,300}$	42 700	8 320	2 190	1 000	510	220	72
	$\eta_{1\,400}$	17 800	3 890	1 120	580	300	140	52
	$\eta_{1\,500}$	8 320	2 040	710	400	200	100	41
MgO–SiO_2	p	34.8	35.5	36.2	40.1	41.4		
Ref. **16**	$\eta_{1\,650}$	610	460	350	250	190		
	$\eta_{1\,750}$	360	270	200	150	120		
	$\eta_{1\,800}$	280	210	180	120	110		
Na_3AlF_6–NaF	p	85	90	95	97.5			
Ref. **7**	$\eta_{1\,000}$	4.8	5.9	6.4	6.7			

(*continued*)

Table 9.9 VISCOSITY OF MOLTEN BINARY SALT SYSTEMS AND OTHER MIXED IONIC MELTS—*continued*

$NaCl–NaNO_3$	p	1.4	2.8	4.2	5.7	7.1	8.6	10.1
Ref. **19**	η_{310}	3.20	3.27	3.68	—	—	—	—
	η_{350}	2.57	2.67	2.79	2.83	3.00	—	—
	η_{400}	2.06	2.21	2.32	2.18	2.11	2.20	2.39
	η_{450}	1.72	1.91	1.98	1.96	1.94	1.73	2.18
$Na_2O–P_2O_5$	p	30.4	32.0	36.0	39.6	43.3	46.5	48.4
Ref. **20**	a	−0.84	−0.63	−0.26	0.11	0.38	0.58	0.59
	$10^{-3}b$	3.57	3.19	2.45	1.87	1.45	1.10	1.00
	r	741–1013	707–1001	657–1017	822–1002	942–1064	1030–1077	995–1 070
$Na_2O–SiO_2$	p	15.2	20.0	30.8	36.9	40.9	45.2	50.0
Ref. **16, 8, 4, 1**	η_{900}	—	4.6×10^6	2.4×10^5	3.2×10^5	1.8×10^5	—	—
	$\eta_{1\,100}$	—	263 000	50 100	26 900	15 100	4070	890
	$\eta_{1\,300}$	118 000	33 100	7 940	4 270	2 340	690	160
	$\eta_{1\,400}$	41 700	16 600	4 900	2 570	1 350	390	100
$PbBr_2–PbCl_2$	p	0	28.4	63.1	80.2	100		
Ref. **12**	η_{450}	—	—	5.53	5.24	4.83		
	η_{500}	4.56	4.45	4.03	3.94	3.73		
	η_{550}	3.54	3.38	3.14	3.05	2.97		
$SiO_2–SrO$	p	36.4	42.0	46.0	57.8	62.9	69.7	
Ref. **16**	$\eta_{1\,650}$	47	360	620	1 380	4 220	—	
	$\eta_{1\,750}$	160	210	370	780	2 240	5 820	
	$\eta_{1\,800}$	130	180	300	580	1 660	4 300	

* *See also* 'Physical Properties Data Compilations Relevant to Energy Storage II Molten Salts: Data on Single and Multi-Component Salt Systems', Janz *et al.*, NSRDS–NBS, 61.

REFERENCES TO TABLE 9.9

1. C. L. Babcock, *J. Am. Ceram. Soc.*, 1934, **17**, 329.
2. S. Karpachev and A. Stromberg, *Z. Anorg. Allg. Chem.*, 1935, **222**, 78.
3. J. R. Rait and R. Hay, *J. R. Tech. Coll. (Glasgow)*, 1938, **4**, 252.
4. E. Preston, *J. Soc. Glass Tech.*, 1938, **22**, 45.
5. V. P. Barzakovskii, *Bull. Acad. Sci. URSS, Classe, Sci. Chim.*, 1940, 825.
6. H. Bloom, B. S. Harrap and E. Heymann, *Proc. R. Soc.*, 1948, **A194**, 237.
7. A. Vajna, *Alluminio*, 1950, **19**, 133.
8. L. Shartsis, S. Spinner and W. Capps, *J. Am. Ceram. Soc.*, 1952, **35**, 155.
9. L. Shartsis, W. Capps and S. Spinner, *ibid.*, 1953, **36**, 319.
10. L. Shartsis, S. Spinner, and H. F. Shermer, *ibid.*, 1954, **37**, 544.
11. J. O'M. Bockris and D. C. Lowe, *Proc. R. Soc.*, 1954, **A226**, 1167.
12. B. S. Harrap and E. Heymann, *Trans. Faraday Soc.*, 1955, **51**, 259.
13. B. S. Harrap and E. Heymann, *ibid.*, 1955, **51**, 268.
14. V. D. Polyakov, *Izvest. Sekt. Fiziko-Khim. Anal.*, 1955, **26**, 147.
15. V. D. Polyakov, *ibid.*, 1955, **26**,191.
16. J. O'M. Bockris, J. D. Mackenzie and J. A. Kitchener, *Trans. Faraday Soc.*, 1955, **51**, 1734.
17. N. P. Luzhnaya, N. N. Evseeva and I. P. Vereshchetina, *Zh. Neorg. Khim.*, 1956, **1**, 1490.
18. G. G. Nozadze, *Soobshch. Akad. Nauk. Gruzin. SSSR*, 1957, **19**, 567.
19. I. P. Vereshchetina and N. P. Luzhnaya, *Izvest. Sekt. Fiziko-Khim. Anal.*, 1954, **25**, 188.
20. C. F. Callis, J. R. Van Wazer and J. S. Metcalf, *J. Am. Chem. Soc.*, 1955, **77**, 1471.
21. G. J. Janz, 'Molten Salts Handbook', Academic Press, London, 1967.